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Kolmogorov and divergent Fourier series

P.L. Ul'yanov

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§1. Introduction

Kolmogorov published about ten papers on the theory of trigonometric and orthogonal series. In fact, each of these papers was the beginning of much research that is still actively pursued today. There is a large body of work due to other authors that is connected with the research of Kolmogorov in this field. We do not even venture a guess at the number.

In this paper, we present just three theorems due to Kolmogorov, we discuss their influence on later research, and in passing we state some unsolved problems.

The material of the paper is organized as follows.

In §2 we give the definitions and auxiliary results that are used in later sections.

In §3 we present Kolmogorov's famous example [1] of a *Fourier series that is almost everywhere divergent*, which led to a new and extensive field of research in the theory of trigonometric series. This example was one of the first results due to Kolmogorov; he obtained it in 1922, when he was a 19 year old student. This truly surprising example still highlights the depth of his ideas and his geometrical intuition.

Nowadays, every thorough treatment of the theory of trigonometric series is bound to include Kolmogorov's example. However, in the well-known monograph of Zygmund ([8], vol. 1) and Hardy and Rogosinski ([10])

almost everywhere divergent Fourier series are constructed by using Fejér polynomials, a method also first used by Kolmogorov [5] in 1926. In Nina Bari's book ([7], vol. 1, 430–443) both the example in Kolmogorov's fundamental paper [1], and Marcinkiewicz's example are presented. In our opinion, some of the clarity and geometric insight in Kolmogorov's original example [1] is lost in these books.

In view of this we have thought it appropriate to give a fairly detailed description of this example, in the form in which it was first proposed by Kolmogorov in 1922.

§4 deals with later results on the divergence of Fourier series.

In 1926 Kolmogorov announced in [6] the theorem that *there exists a trigonometric Fourier series of class L^2 which after an appropriate rearrangement of the terms is almost everywhere divergent.* This result has a somewhat dramatic history. For Kolmogorov delayed the publication of a detailed proof until a more convenient time. However, as time passed, his interests changed, and the proof did not appear. Thirty-five years went by, and then in 1960 a brief outline of the construction of the required example was given in a paper by the Polish mathematician Zagorski [34]. Immediately afterwards in 1960 we proved in [36] an analogue of Kolmogorov's theorem for Walsh and Haar systems(1) and we also raised the question of an analogue for arbitrary orthonormal complete (ONC) systems. In 1961, both Olevskii [41] and the present author [37] and [38] used the theorem on series of Haar systems mentioned above to prove the corresponding result for ONC systems. Furthermore, we were able to show in these papers that there is a corresponding result for any bases in L^2. After this, a particularly active area was the question of unconditional convergence almost everywhere. In §5 we present results on the unconditional convergence of series in the Haar system and in bases, and some further developments.

In §6 we first state the Kolmogorov-Men'shov theorem: *there exists a Fourier series of class L^2 in some orthonormal system $\{\varphi_n\}$ with $|\varphi_n(t)| \equiv 1$ for $t \in [0, 1]$ that is everywhere divergent on the interval* $[0, 1]$.

A particular consequence of this theorem is the fundamental fact that not every totally bounded ON system is a system of convergence almost everywhere.

This is one of the main results proved in 1926 by Kolmogorov and Men'shov [6]. In the same article they proved that if $\omega(n) = o(\log n)$, then the sequence $\{\omega(n)\}$ is not a Weyl multiplier for almost everywhere convergence of series in some ON system of sign functions(2) $\{\varphi_n(t)\}_{n=1}^{\infty}$. They also mention ([6], 432) that it was an open question whether this result was best possible. The answer to this question came in 1975, when

(1)We are concerned with series in these systems, of course.

(2)That is, the $\varphi_n(t)$ take only the values 1 and −1.

Kashin [67] constructed an ON system $\{\varphi_n(t)\}$ such that $|\varphi_n(t)| \equiv 1$, but any sequence $\{\omega(n)\}$ with $\omega(n) = o(\log^2 n)$ is not a Weyl multiplier for almost everywhere convergence of series in the system $\{\varphi_n\}$. §6 completes the proof of this result (we make use of some simplifications due to Tandori [68]).

§2. Definitions and auxiliary results

1. A set $E \subset [a, b]$ is said to be of type F_σ if it can be represented in the form $E = \bigcup_{n=1}^{\infty} E_n$, where E_n is a closed set in $[a, b]$. A set $E \subset [a, b]$ is of *type* G_δ if

$$E = [a, b] \setminus A,$$

where A is a set of type F_σ (or, if $E = \bigcap_{n=1}^{\infty} G_n$, where the G_n are open sets in $[a, b]$).

Similarly, E is of type $F_{\sigma\delta}$ ($G_{\delta\sigma}$) if $E = \bigcap_{n=1}^{\infty} B_n$ ($E = \bigcup_{n=1}^{\infty} B_n$).

Suppose that $f_n(t)$ is a given sequence of functions defined on $[a, b]$. Then the set $E_d = \{t : t \in [a, b], \lim_{n\to\infty} f_n(t)$ does not exist$\}$ is called the *set of divergence* of the sequence $\{f_n\}$.

The set

$$E_c = [a, b] \setminus E_d$$

is the *set of convergence* of $\{f_n\}$.

By analogy, the set

$$E_{ud} = \{t\colon t \in [a, b], \overline{\lim_{n\to\infty}} |f_n(t)| = \infty\} \quad (E_{bd} = E_d \setminus E_{ud})$$

[illegible] *unbounded (bounded)* divergence of the sequence

It is well known (see [24] English ed., 303–311) that if the $f_n(t)$ are continuous on $[a, b]$ for $n = 1, 2, \ldots$, then E_c is an $F_{\sigma\delta}$ set, E_d a $G_{\delta\sigma}$ set, E_{ud} a G_δ set, and E_{bd} a $G_{\delta\sigma}$ set, and these results cannot be sharpened.

The definitions for the sets of convergence, divergence, and so on for series are completely analogous (in this case we have to consider the behaviour of the partial sums of these series).

2. Let f be a function of period 2π such that $f \in L^p(0, 2\pi)$ for some $p \in [1, \infty)$ (if $p = \infty$, we suppose that f is continuous). Then the function

$$\omega_p(\delta, f) = \sup_{|h|\leqslant\delta} \left\{\int_0^{2\pi} |f(t+h) - f(t)|^p\, dt\right\}^{1/p} \text{ for } 1 \leqslant p < \infty$$

and

$$\omega(\delta, f) = \omega_\infty(\delta, f) = \sup_{\substack{|h|\leqslant\delta \\ t\in[0, 2\pi]}} |f(t+h) - f(t)| \quad \text{for } p = \infty$$

is called the modulus of continuity of f in the space $L^p(0, 2\pi)$.

The definition of the modulus of continuity when f is not periodic is analogous. For example, if $f \in L^p(0, 1)$ for $1 \leqslant p < \infty$, then

$$\omega_p(\delta, f) = \sup_{0 \leqslant h \leqslant \delta} \left\{ \int_0^{1-h} |f(t+h) - f(t)|^p \, dt \right\}^{1/p}.$$

3. The following theorem due to Beppo Levi (see, for example [75]) is often used in the theory of functions.

If $f_n(t) \in L(E)$ are non-negative functions and

$$\sum_{n=1}^{\infty} \int_E f_n(t)\, dt < \infty, \tag{2.1}$$

then the series

$$\sum_{n=1}^{\infty} f_n(t)$$

converges almost everywhere on E to a function $f \in L(E)$. Moreover, the series (2.1) *is also convergent to f in the norm of $L(E)$.*

4. Let $\{\varphi_n(t)\}$ be an orthonormal system (ONS) on $[a, b]$. The orthogonal series

$$\sum_{n=1}^{\infty} a_n \varphi_n(t) \tag{2.2}$$

is called a *Fourier series of class L^p* with $1 \leqslant p \leqslant \infty$ if there is a function $F \in L^p(a, b)$ such that (2.2) is the Fourier series of F in the system $\{\varphi_n\}$. In this case the partial sums of (2.2) are denoted by $S_n(t, F)$ (or $s_n(t, F)$) and the coefficients by $a_n(F)$, and so on.

A non-negative non-decreasing sequence $\{\omega(n)\}_{n=1}^{\infty}$ is called a *Weyl multiplier for convergence almost everywhere* (a.e.) (for *unconditional convergence almost everywhere*) of series in the system $\{\varphi_n\}$ if from

$$\sum_{n=1}^{\infty} a_n^2 \omega(n) < \infty$$

it follows that (2.2) converges a.e. on $[a, b]$ (respectively, (2.2) converges after any rearrangement)[1] a.e. on $[a, b]$). In this case we write

$$\{\varphi_n(t)\} \in W(c, E, \omega(n)), \quad \{\varphi_n(t)\} \in W(uc, E, \omega(n)) \text{ with } E = [a, b].$$

If it is clear from the context what the set E is, then it will be omitted.

If $\{\varphi_n\}$ is such that for $\omega(n) \equiv 1$ the system $\{\varphi_n\} \in W(c, 1)$ (or $\{\varphi_n\} \in W(uc, 1)$), then $\{\varphi_n\}$ is called a *system of convergence a.e.* (a *system of unconditional convergence a.e.*).

[1]The exceptional set of divergence (of measure zero) depends, in general, on the rearrangement.

If $\{\varphi_n\} \in W(c, \omega(n))$, but for any sequence $\gamma(n) = o(\omega(n))$ $\{\varphi_n\} \notin$ $\notin W(c, \gamma(n))$ as $n \to \infty$, then $\{\omega(n)\}$ is said to be an *exact* Weyl multiplier for convergence a.e. on $[a, b]$ for series of the form (2.2).

The definition of an *exact* Weyl multiplier for unconditional convergence a.e. is analogous.

We recall the standard theorem of Men'shov and Rademacher (see [9], 162) that[1] for any ONS

$$\{\varphi_n\} \in W(c, \log^2(n+1)).$$

In addition, Men'shov ([74]; see also [9], 167) has shown (and this is quite important) that this result is best possible in the whole class of ONS.

As regards unconditionally convergent series of the form (2.2), Orlicz [81] (see also [9], 170-171) has shown the following:

For any ONS $\{\varphi_n\}$

$$\{\varphi_n\} \in W(uc, \omega(n)),$$

if $\omega(n) = \tau(n)\log^2(n+1)$, where $\tau(n) \uparrow \infty$ and

$$\sum_{k=1}^{\infty} \frac{1}{\tau(n_k)} < \infty, \quad \log n_{k+1} = O(\log n_k) \quad \text{as} \quad k \to \infty$$

for some sequence of natural numbers $n_k \uparrow \infty$.

Tandori [76] has shown that this theorem is in a certain sense best possible.

5. The sequence of trigonometric functions $\{\cos nt, \sin nt\}$ is, of course, orthogonal on the interval $[0, 2\pi]$. If $f(t) \in L(0, 2\pi)$ is a given function,

$$f(t) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nt + b_n \sin nt) \tag{2.3}$$

have the form ([7], 95-96)

$$S_m(t, f) = \frac{1}{\pi} \int_0^{2\pi} f(y+t) D_m(y)\, dy, \tag{2.4}$$

with the Dirichlet kernel

$$D_m(y) = \frac{1}{2} + \cos y + \ldots + \cos my = \frac{\sin\left(m+\frac{1}{2}\right) y}{2 \sin \frac{y}{2}}.$$

It is clear that (see (2.4))

$$|S_m(t, f)| \leqslant \frac{2m+1}{2\pi} \int_0^{2\pi} |f(y)|\, dy \quad \text{for all } t \in [0, 2\pi] \text{ and } m = 0, 1, \ldots$$

[1]By $\log a$ we mean the logarithm of a to the base 2.

***Lemma* 2.1.** *If a 2π-periodic function f is of bounded variation on $[0, 2\pi]$, then*

$$|S_m(t, f)| \leqslant \sup_{y\in[0,\,2\pi]} |f(y)| + 2\bigvee_0^{2\pi} f \quad \textit{for all } t\in[0, 2\pi] \textit{ and } m=0, 1, \ldots$$

A proof is given in [7].

***Theorem* 2.1** (Jordan; see [7], 114–115). *If $f \in V(0, 2\pi)$, then*

$$\lim_{m\to\infty} S_m(t, f) = \frac{f(t+0)+f(t-0)}{2} \quad \textit{for all } t\in(-\infty, \infty).$$

If f is continuous on (a, b), then $S_m(t, f)$ converges to $f(t)$, uniformly on any interval $[c, d] \subset (a, b)$.

In what follows, if we are given a trigonometric series of the form (2.3), then the series

$$\sum_{n=1}^{\infty} (a_n \sin nt - b_n \cos nt)$$

is said to be *conjugate* to (2.3).

6. Suppose that r_0 is the function $r_0(t) = 1$ for $0 < t < 1/2$, $r_0(t) = -1$ for $1/2 < t < 1$, and $r_0(t) = 0$ for $t = 0, 1/2$. We extend $r_0(t)$ to the real line so that it has the period 1, and we set $r_n(t) = r_0(2^n t)$. Then $\{r_n(t)\}_{n=0}^{\infty}$ is called the *Rademacher system.* This is an ON system of convergence almost everywhere (see [9], 125). Furthermore, Steinhaus [77] has shown that it is a system of unconditional convergence a.e.

We also recall the definition of the ON Haar system $\{\chi_m(t)\}_{m=1}^{\infty}$ for $t \in [0, 1]$. We set

$$\chi_1(t) = \chi_0^{(0)}(t) \equiv 1$$

for $t \in [0, 1]$, and for $m = 2^n + k$ with $1 \leqslant k \leqslant 2^n$ and $n = 0, 1, \ldots$

$$\chi_m(t) = \chi_n^{(k)}(t) = \begin{cases} \sqrt{2^n} & \text{for } t\in\left(\frac{2k-2}{2^{n+1}}, \frac{2k-1}{2^{n+1}}\right), \\ -\sqrt{2^n} & \text{for } t\in\left(\frac{2k-1}{2^{n+1}}, \frac{2k}{2^{n+1}}\right), \\ 0 & \text{for the remaining } t \text{ in } [0, 1]. \end{cases}$$

The point $t_m = (2k-1)/2^{n+1}$ is called an essential zero of the function χ_m for $m \geqslant 2$. We extend these functions with period 1 to the whole real line. As we have defined it, this system $\{\chi_m\}$ differs from the standard Haar system (see [11], English ed., 46) only at a countable set of points. This clearly does not matter if in the problems we consider we neglect sets of measure zero (as we often do).

Haar has shown (see [11], English ed., 47–50) that if $f(t) \in L(0, 1)$, then the Fourier-Haar series

$$\sum_{m=1}^{\infty} a_m(f)\chi_m(t) \quad \text{with} \quad a_m(f) = \int_0^1 f(y)\chi_m(y)\,dy$$

converges almost everywhere on $[0, 1]$ to $f(t)$. Thus, $\{\chi_m\}$ is a system of convergence a.e.

The Haar and Rademacher functions are connected by

$$\sqrt{2^n}\, r_n(t) = \sum_{k=1}^{2^n} \chi_n^{(k)}(t) \quad \text{for} \quad t \in (-\infty, \infty) \text{ and } n = 0, 1, \ldots$$

7. A generalization of the concept of an ONS in the space $L^2(0, 1)$ is the notion of a basis. A set of functions $\{f_n\}_{n=1}^{\infty}$ with $f_n \in L^2(0, 1)$ is called a *basis* in $L^2(0, 1)$ if for every function $f \in L^2(0, 1)$ there is a unique sequence of numbers $\{a_n\}$ such that the series

$$\sum_{n=1}^{\infty} a_n f_n(t) \tag{2.5}$$

converges to f in the $L^2(0, 1)$-norm.

Since this is independent of the norm, we may assume that $\|f_n\|_2 = 1$ for $n = 1, 2, \ldots$

A basis set satisfies Banach's theorem.

Banach's theorem ([48]). *If $\{f_n\}_{n=1}^{\infty}$ is a basis in $L^2(0, 1)$, then there is a system of functions $\{g_m\}_{m=1}^{\infty}$ such that $g_m \in L^2(0, 1)$ and*

$$\int_0^1 f_n(t)\, g_m(t)\, dt = \begin{cases} 0 & \text{for } n \neq m, \\ 1 & \text{for } n = m, \end{cases}$$

that is, $\{f_n, g_m\}$ is a biorthonormal set on $[0, 1]$, and $\{g_m\}$ is dual to the system $\{f_n\}$.

[illegible] $\int_0^1$ [illegible]

$n = 1, 2, \ldots$ The next result is almost obvious ([38], 49).

Lemma 2.2. *If $\{f_n\}$ is a basis in $L^2(0, 1)$ and a series of the form (2.5) is such that for some sequence of natural numbers $n_k \uparrow \infty$ the partial sums*

$$\sum_{n=1}^{n_k} a_n f_n$$

converge in $L^2(0, 1)$ to some function F, then (2.5) is the expansion of F in the basis $\{f_n\}$.

§3. Kolmogorov's example of a trigonometric Fourier series that diverges almost everywhere

In this section we give a detailed account of the example. We use the same method of proof as Kolmogorov [1]. If our attention to the finer details obscures the main ideas, we refer the reader to the original paper [1].

At the basis of the proof lies a lemma due to Kolmogorov.

***Lemma* 3.1.** *There exists a sequence of 2π-periodic functions $\{\varphi_n(t)\}_{n=1}^{\infty}$ such that*

1°) $\varphi_n(t) \geqslant 0$ *for* $t \in [0, 2\pi]$, *and*

$$\int_0^{2\pi} \varphi_n(t)\,dt = 2 \qquad (n = 1, 2, \ldots); \tag{3.1}$$

2°) *the functions $\varphi_n(t)$ are of bounded variation on* $[0, 2\pi]$;

3°) *with each function φ_n we can associate a positive number M_n, a set $E_n \subset [0, 2\pi]$, and an integer q_n such that*

$$\lim_{n\to\infty} M_n = \infty; \tag{3.2}$$

$$\lim_{n\to\infty} \operatorname{mes} E_n = 2\pi. \tag{3.3}$$

For each point t of E_n there is a number $p_n(t)$ such that

$$p_n(t) \leqslant q_n \quad u \quad |S_{p_n(t)}(t, \varphi_n)| > M_n \qquad (n = 1, 2, \ldots), \tag{3.4}$$

where $S_p(t, \varphi_n)$ is the partial sum of the trigonometric Fourier series of φ_n at t.

Lemma 3.1 states that there exist functions of bounded variation whose $L(0, 2\pi)$-norm is 2, but for which some partial sums of the Fourier series are arbitrarily large at the points of a set of measure close to 2π (see (3.1)–(3.4)).

A consequence of this is the next theorem.

***Theorem* 3.1** (Kolmogorov [1]). *There exists a function $\Phi(t) \in L(0, 2\pi)$ such that its trigonometric Fourier series diverges almost everywhere.*

Proof. By induction we can find an increasing sequence of natural numbers $n_1 < n_2 < \ldots < n_k$ such that

$$\text{A)} \quad \frac{1}{\sqrt{M_{n_k}}} < \frac{1}{2^k} \quad \text{for } k = 1, 2, \ldots; \tag{3.5}$$

B) if

$$B_n = \sup_{\substack{m \geqslant 0 \\ t \in [0, 2\pi]}} |S_m(t, \varphi_n)|, \tag{3.6}$$

then

$$\sum_{i=1}^{k-1} B_{n_i} < \frac{1}{2}\sqrt{M_{n_k}} \qquad (k = 1, 2, \ldots);$$

$$\text{C)} \quad 2q_{n_i} + 1 \leqslant \frac{\sqrt{M_{n_k}}}{2^k} \quad \text{for all} \quad i < k. \tag{3.7}$$

Since the functions φ_n are of bounded variation, by Lemma 2.1 the numbers B_n are finite. Therefore, if we have already chosen $n_1, n_2, \ldots, n_{k-1}$ then by taking $n_k > n_{k-1}$ sufficiently large, we can satisfy (3.5)–(3.7), because $M_n \to \infty$ as $n \to \infty$ (see (3.2)).

We put

$$\Phi(t) = \sum_{i=1}^{\infty} \frac{1}{\sqrt{M_{n_i}}} \varphi_{n_i}(t). \tag{3.8}$$

Since $\varphi_n(t) \geqslant 0$ and (see (3.1) and (3.5))

$$\sum_{i=1}^{\infty} \frac{1}{\sqrt{M_{n_i}}} \int_0^{2\pi} \varphi_{n_i}(t)\, dt = \sum_{i=1}^{\infty} \frac{2}{\sqrt{M_{n_i}}} < 2, \tag{3.9}$$

by Beppo Levi's theorem (see § 2) the series (3.8) converges a.e. on $[0, 2\pi]$ to some function $\Phi(t)$ and $\Phi \in L(0, 2\pi)$. Moreover, (3.8) converges to $\Phi(t)$ in the $L(0, 2\pi)$-norm. This follows from (3.9). Hence, for any natural number N the Fourier coefficient

$$a_N(\Phi) = \frac{1}{\pi} \int_0^{2\pi} \Phi(t) \cos Nt\, dt = \sum_{i=1}^{\infty} \frac{1}{\sqrt{M_{n_i}}} a_N(\varphi_{n_i})$$

and $b_N(\Phi)$ has a similar expression. Thus, the Fourier coefficients of Φ are equal to the sum of the Fourier coefficients of the functions

$$\frac{1}{\sqrt{M_{n_i}}} \varphi_{n_i}(t) \qquad (i = 1, 2, \ldots).$$

Thus, the partial sums are

$$S_N(t, \Phi) = \sum_{i=1}^{\infty} \frac{1}{\sqrt{M_{n_i}}} S_N(t, \varphi_{n_i}) \qquad (N = 1, 2, \ldots). \tag{3.10}$$

We choose a natural number k and a point $t \in E_{n_k}$, and we consider the [illegible] $N = p_{n_k}(t)$. Clearly (see (3.10))

$$|S_{p_{n_k}(t)}(t, \Phi)| = \Big| \sum_{i=1}^{\infty} \frac{1}{\sqrt{M_{n_i}}} S_{p_{n_k}(t)}(t, \varphi_{n_i}) \Big| \geqslant \tag{3.11}$$

$$\geqslant \frac{1}{\sqrt{M_{n_k}}} |S_{p_{n_k}(t)}(t, \varphi_{n_k})| - I_1 - I_2,$$

where (see (3.6))

$$I_1 = \sum_{i=1}^{k-1} \frac{1}{\sqrt{M_{n_i}}} |S_{p_{n_k}(t)}(t, \varphi_{n_i})| \leqslant \tag{3.12}$$

$$\leqslant \sum_{i=1}^{k-1} \frac{1}{\sqrt{M_{n_i}}} \sup_{\substack{1 \leqslant p < \infty \\ t \in [0, 2\pi]}} |S_p(t, \varphi_{n_i})| \leqslant \sum_{i=1}^{k-1} \frac{B_{n_i}}{\sqrt{M_{n_i}}} < \frac{1}{2}\sqrt{M_{n_i}},$$

and

$$I_2 = \sum_{i=k+1}^{\infty} \frac{1}{\sqrt{M_{n_i}}} |S_{p_{n_k}(t)}(t, \varphi_{n_i})|.$$

Since always (see § 2)

$$|S_p(t, f)| \leqslant \left(\frac{1}{2} + p\right) \frac{1}{\pi} \int_0^{2\pi} |f(y)|\, dy,$$

we have (see (3.4), (3.1), and (3.7))

$$(3.13)\qquad I_2 \leqslant \sum_{i=k+1}^{\infty} \frac{1}{\sqrt{M_{n_i}}}\left(\frac{1}{2}+q_{n_k}\right)\frac{1}{\pi}\int_0^{2\pi} |\varphi_{n_i}(y)|\,dy \leqslant$$

$$\leqslant \frac{1}{\pi}\sum_{i=k+1}^{\infty}\frac{2q_{n_k}+1}{\sqrt{M_{n_i}}} \leqslant \sum_{i=k+1}^{\infty}\frac{1}{2^i}=\frac{1}{2^k}.$$

From (3.4) and (3.11)-(3.13) we find that

$$(3.14)\quad |S_{p_{n_k}}(t,\Phi)| > \sqrt{M_{n_k}}-\frac{1}{2}\sqrt{M_{n_k}}-\frac{1}{2^k}=\frac{1}{2}\sqrt{M_{n_k}}-\frac{1}{2}\quad \text{for } t\in E_{n_k}.$$

This inequality states that at points of the set E_{n_k} the main contribution to the partial sums of the Fourier series of the function Φ is given by

$$\frac{1}{\sqrt{M_{n_k}}}\varphi_{n_k}(t).$$

For this function determines the behaviour of the Fourier series of Φ at points of E_{n_k}. This can also be seen without a detailed calculation (3.14) if we look carefully at the conditions A), B), and C), which can be satisfied according to the basic Lemma 3.1.

Now we put

$$(3.15)\qquad E=\overline{\lim_{k\to\infty}}\, E_{n_k}.$$

Since the limit superior is

$$(3.16)\qquad E=\bigcap_{j=1}^{\infty}\bigcup_{k=j}^{\infty} E_{n_k},$$

we have (see 3°)

$$2\pi \geqslant \operatorname{mes}\bigcup_{k=j}^{\infty} E_{n_k} \geqslant \operatorname{mes} E_{n_k}\quad \text{for any}\quad k\geqslant j,$$

and therefore (see (3.3))

$$\operatorname{mes}\bigcup_{k=j}^{\infty} E_{n_k}=2\pi.$$

This implies (see (3.15) and (3.16)) that mes $E = 2\pi$.

Let us verify that the Fourier series of $\Phi(t)$ diverges at every point of E. If $t \in E$, then $t \in E_{n_k}$ for infinitely many k. But then it follows from (3.14) (see also (3.2)) that

$$(3.17)\qquad \overline{\lim_{m\to\infty}}\, |S_m(t,\Phi)| = \infty.$$

Since mes $E = 2\pi$, this completes the proof of Theorem 3.1.

Remark 3.1. From (3.17) it is clear that the Fourier series of $\Phi(t)$ diverges to infinity almost everywhere on $[0, 2\pi]$.

Now we have to check the validity of Lemma 3.1.

Proof of Lemma 3.1. Let $n \geqslant n_0$ be a natural number, where n_0 is sufficiently large. We construct an increasing sequence of odd numbers

$$\lambda_1 = 1,\ \lambda_2,\ \ldots,\ \lambda_n,$$

on which we impose below some additional conditions. We note that for every n we have constructed a finite sequence $\{\lambda_k(n)\}_{k=1}^{n}$.

We define a sequence of natural numbers $m_1, m_2, \ldots, m_n$ such that

$$m_1 = n,\ 2m_k + 1 = \lambda_k(2n+1) \text{ with } 1 \leqslant k \leqslant n. \tag{3.18}$$

We set

$$A_k = \frac{4\pi}{2n+1}\, k \quad \text{for } 1 \leqslant k \leqslant n,$$

and take intervals

$$\Delta_k = \left[A_k - \frac{1}{m_k^2},\ A_k + \frac{1}{m_k^2}\right] \quad \text{for } 1 \leqslant k \leqslant n.$$

The Δ_k are disjoint and lie in $[0, 2\pi]$ because (see (3.18))

$$\frac{1}{m_k^2} < \frac{2\pi}{2n+1}.$$

We define a function $\varphi_n(t)$ by putting

$$\varphi_n(t) = \begin{cases} \dfrac{m_k^2}{n} & \text{for } t \in \Delta_k \text{ with } 1 \leqslant k \leqslant n, \\ 0 & \text{for } t \in [0, 2\pi] - \bigcup\limits_{k=1}^{n} \Delta_k, \\ \varphi_n(2\pi + t) & \text{for all } t \in (-\infty, \infty). \end{cases} \tag{3.19}$$

At the k-th "step" we define

$$\varphi_n^{(k)}(t) = \begin{cases} \varphi_n(t) & \text{for } t \in \Delta_k \\ 0 & \text{for } t \in [0, 2\pi] - \Delta_k. \end{cases} \quad (1 \leqslant k \leqslant n)$$

It is clear that (see (3.19))

$$\varphi_n(t) = \sum_{k=1}^{n} \varphi_n^{(k)}(t),$$

that $\varphi_n(t)$ is non-negative, and that the integral

$$\int_0^{2\pi} \varphi_n(t)\, dt = \sum_{k=1}^{n} \frac{m_k^2}{n} \frac{2}{m_k^2} = 2,$$

that is, (3.1) holds. It is also clear that φ_n is of bounded variation on $[0, 2\pi]$, that is, Lemma 3.1, 2° holds.

We take intervals

$$\sigma_k = \left[A_{k-1} + \frac{2}{n^2},\ A_k - \frac{2}{n^2}\right] \quad \text{for } 2 \leqslant k \leqslant n,$$

which makes sense, since

$$A_{k-1} + \frac{2}{n^2} < A_k - \frac{2}{n^2} \quad \text{for } 2 \leqslant k \leqslant n.$$

These intervals σ_k $(2 \leqslant k \leqslant n)$ are disjoint from the Δ_i $(1 \leqslant i \leqslant n)$, because

$$\frac{1}{m_{k-1}^2} < \frac{2}{n^2} \quad \text{for} \quad 2 \leqslant k \leqslant n.$$

Now we draw attention to the fact that the interval σ_k "almost completely" fills $[A_{k-1}, A_k]$, that is,

$$\operatorname{mes} \sigma_k = \operatorname{mes} [A_{k-1}, A_k] - \frac{4}{n^2} = \frac{4\pi}{2n+1} - \frac{4}{n^2},$$

whose union almost gives the entire interval $[0, 2\pi]$.

Clearly,

$$S_{m_k}(x, \varphi_n) = \frac{1}{\pi} \int_0^{2\pi} \varphi_n(t) D_{m_k}(t-x)\, dt = \frac{1}{\pi} \int_0^{2\pi} \varphi_n(t) \frac{\sin\left(m_k + \frac{1}{2}\right)(t-x)}{2 \sin \frac{1}{2}(t-x)} dt.$$

We note that we have not yet chosen the numbers λ_k for $2 \leqslant k \leqslant n$. By (3.19),

$$(3.20) \quad S_{m_k}(x, \varphi_n) = \frac{1}{\pi} \sum_{i=1}^{k-1} \int_{\Delta_i} \varphi_n(t) D_{m_k}(t-x)\, dt + \\ + \frac{1}{\pi} \sum_{i=k}^{n} \int_{\Delta_i} \varphi_n(t) D_{m_k}(t-x)\, dt = J_1(x) + J_2(x).$$

Now

$$(3.21) \quad \frac{1}{\pi} \int_{\Delta_i} \varphi_n(t) D_{m_k}(t-x)\, dt = \\ = \frac{1}{\pi} \int_0^{2\pi} \varphi_n^{(i)}(t) D_{m_k}(t-x)\, dt = S_{m_k}(x, \varphi_n^{(i)}).$$

We assume that the numbers $\lambda_1 = 1, \lambda_2, \ldots, \lambda_{k-1}$ have already been chosen (that is, we have chosen $m_1 = n, m_2, \ldots, m_{k-1}$). Next we choose m_k.

We consider the partial sums

$$S_{m_k}(x, \varphi_n^{(i)}) \quad \text{for} \quad x \in \sigma_k \quad \text{and} \quad 1 \leqslant i \leqslant k-1.$$

Since the Δ_i for $1 \leqslant i \leqslant k-1$ and the σ_k are disjoint, and since $\varphi_n^{(i)}$ is of bounded variation on $[0, 2\pi]$, we have (see Theorem 2.1)

$$S_m(x, \varphi_n^{(i)}) \to 0 \quad \text{as} \quad m \to \infty, \quad x \in \sigma_k \quad \text{and} \quad 1 \leqslant i \leqslant k-1.$$

This convergence is uniform on σ_k. Thus, we can choose m_k such that $2m_k + 1 = \lambda_k(2n+1)$, where $\lambda_k > \lambda_{k-1}$, λ_k is odd and

$$(3.22) \qquad |S_{m_k}(x, \varphi_n^{(i)})| < \frac{1}{n} \quad \text{for} \quad x \in \sigma_k \quad \text{and} \quad 1 \leqslant i \leqslant k-1.$$

We note that (3.22) can also be obtained without reference to Theorem 2.1. For the partial sum (see (3.21) and (3.19)) can be written as an integral

$$S_{m_k}(x, \varphi_n^{(i)}) = \frac{m_i^2}{2\pi n} \int_{\Delta_i} \frac{1}{\sin \frac{1}{2}(t-x)} \sin\left(m_k + \frac{1}{2}\right)(t-x)\, dt \qquad (i < k),$$

to which we can apply the second mean value theorem. If we take into account that the function

$$\frac{1}{|\sin\frac{1}{2}(t-x)|} \quad \text{for} \quad x\in\sigma_k \quad \text{and} \quad t\in\Delta_i \quad \text{with} \quad 1\leqslant i\leqslant k-1$$

is bounded, then we obtain (3.22) for sufficiently large m_k.

Thus, we have chosen m_k in accordance with (3.22). And so the m_k are defined for all k, $1\leqslant k\leqslant n$, and the definition of $\varphi_n(t)$ is complete.

What does this choice of the m_k imply?

Firstly, from (3.22) it follows at once (see (3.20)) that

$$|J_1(x)|<1 \quad \text{for} \quad x\in\sigma_k. \tag{3.23}$$

Once again, we emphasize that this inequality holds at the expense of taking the m_k large (compared with $m_1<m_2<\ldots<m_{k-1}$).

Secondly, we can estimate the terms in the sum $J_2(x)$ (see (3.20) and (3.21)), that is, the partial sums $S_{m_k}(x,\varphi_n^{(i)})$ for $x\in\sigma_k$ and $i\geqslant k$.

We note that the points A_i form an arithmetic progression for which

$$\frac{2m_k+1}{2}(A_i-A_k)=\frac{\lambda_k(2n+1)}{2}\frac{4\pi}{2n+1}(i-k)=(i-k)\lambda_k 2\pi,$$

so that

$$\sin\frac{2m_k+1}{2}(A_i-x)=\sin\frac{2m_k+1}{2}(A_k-x). \tag{3.24}$$

We recall that always for the Dirichlet kernel

$$|D_p'(t)|\leqslant\sum^{p}|\nu\sin\nu t|\leqslant p^2 \quad \text{for any} \quad t\in(-\infty,\infty) \tag{3.25}$$

the difference is

$$|A_i-t|\leqslant\frac{1}{m_i^2} \quad \text{for} \quad t\in\Delta_i \tag{3.26}$$

and that

$$0<A_i-x<A_i-A_{k-1}=(i-k+1)\frac{4\pi}{2n+1}<(i-k+1)\frac{2\pi}{n} \tag{3.27}$$

for $i\geqslant k$ and $x\in\sigma_k$.

Next, the partial sums satisfy (see (3.21) and (3.24))

$$S_{m_k}(x,\varphi_n^{(i)})=\frac{1}{\pi}\int_{\Delta_i}\varphi_n(t)D_{m_k}(t-x)\,dt= \tag{3.28}$$

$$=\frac{1}{\pi}\int_{\Delta_i}\varphi_n(t)D_{m_k}(A_i-x)\,dt+\frac{1}{\pi}\int_{\Delta_i}\varphi_n(t)[D_{m_k}(t-x)-D_{m_k}(A_i-x)]\,dt=$$

$$=\frac{\sin\frac{2m_k+1}{2}(A_i-x)}{\pi 2\sin\frac{1}{2}(A_i-x)}\cdot\frac{2}{n}+K=\sin\frac{2m_k+1}{2}(A_k-x)\frac{1}{\pi n\sin\frac{A_i-x}{2}}+K(x).$$

By Lagrange's theorem (see (3.25) and (3.26))

$$|D_{m_k}(t-x)-D_{m_k}(A_i-x)|\leqslant m_k^2|A_i-t|\leqslant 1 \quad \text{for} \quad t\in\Delta_i$$

therefore,

$$|K(x)| \leqslant \frac{1}{\pi} \int_{\Delta_i} \varphi_n(t)\, dt < \frac{1}{n}.$$

As a result we obtain (see (3.28))

$$(3.29)\quad S_{m_k}(x, \varphi_n^{(i)}) = \sin\left(m_k + \frac{1}{2}\right)(A_k - x) \frac{1}{\pi n \sin \frac{A_i - x}{2}} + \frac{\tau}{n} \quad \text{with } |\tau| < 1$$

and (see (3.20) and (3.29))

$$(3.30)\quad J_2(x) = \sin\left(m_k + \frac{1}{2}\right)(A_k - x) \sum_{i=k}^{n} \frac{1}{\pi n \sin \frac{A_i - x}{2}} + \tau,$$

where $|\tau| < 1$.

By (3.27),

$$\sum_{i=k}^{n} \frac{1}{\pi n \sin \frac{A_i - x}{2}} > \frac{n}{\pi^2 n (i-k+1)} = \frac{1}{\pi^2 (i-k+1)} \quad \text{for} \quad x \in \sigma_k,$$

since $0 < \sin u < u$ and $0 < u < \pi$, and so (see (3.30))

$$|J_2(x)| > \frac{\sin\left(m_k + \frac{1}{2}\right)(A_k - x)}{\pi^2} \sum_{j=1}^{n-k+1} \frac{1}{j} - 1 \geqslant$$

$$\geqslant \frac{1}{\pi^2} \left| \sin\left(m_k + \frac{1}{2}\right)(A_k - x) \right| \ln(n-k+2) - 1 \quad \text{for} \quad x \in \sigma_k.$$

Thus (see (3.20) and (3.23))

$$(3.31)\quad |S_{m_k}(x, \varphi_n)| \geqslant \frac{1}{\pi^2} \left| \sin\left(m_k + \frac{1}{2}\right)(A_k - x) \right| \ln(n-k+2) - 2$$

for $x \in \sigma_k$ with $2 \leqslant k \leqslant n$.

This is the main bound. What has it cost to obtain it? Firstly, the partial sums $S_{m_k}(x, \varphi_n^{(i)})$ are small on σ_k for $1 \leqslant i \leqslant k-1$ (see (3.22)). Secondly, for $x \in \sigma_k$ each function $\varphi_n^{(i)}$ (the i-th "step") with $i \geqslant k$ makes a contribution (to the partial sum $S_{m_k}(x, \varphi_n)$) roughly equal to (see (3.29))

$$C \sin\left(m_k + \frac{1}{2}\right)(A_k - x) \frac{1}{i-k+1},$$

where the second factor is a term of the harmonic series. For fixed $x \in \sigma_k$ these contributions have the same sign for $k \leqslant i \leqslant n$, and so the final estimate takes the form (3.31).

Now we can find M_n, q_n, and the set E_n.

In (3.31) we take $k \leqslant n - \sqrt{n}$. Then

$$(3.32)\quad |S_{m_k}(x, \varphi_n)| \geqslant \frac{\left|\sin\left(m_k + \frac{1}{2}\right)(A_k - x)\right|}{2\pi^2} \ln n - 2 \quad \text{for} \quad 2 \leqslant k \leqslant n - \sqrt{n},$$

where $n \geqslant n_0$. We set

$$M_n = \sqrt{\ln n} - 2, \quad q_n = m_n,$$

and

$$E_n = \bigcup_{2 \leqslant k \leqslant n-\sqrt{n}} E_n^{(k)},$$

where

$$E_n^{(k)} = \left\{ x\colon\ x \in \sigma_k \text{ and } \left| \sin\left(m_k + \frac{1}{2}\right)(A_k - x) \right| \geqslant \frac{2\pi^2}{\sqrt{\ln n}} \right\}.$$

If n_0 and m_2 have been chosen large enough, then for $n \geqslant n_0$

$$\operatorname{mes}\left\{x\colon\ x \in [A_{k-1}, A_k],\ \left| \sin\left(m_k + \frac{1}{2}\right)(A_k - x) \right| < \frac{2\pi^2}{\sqrt{\ln n}} \right\} = O\left(\frac{1}{n\sqrt{\ln n}}\right),$$

and

$$\operatorname{mes} E_n^{(k)} = \operatorname{mes} \sigma_k - O\left(\frac{1}{n\sqrt{\ln n}}\right) \quad \text{as} \quad n \to \infty.$$

Therefore,

$$\operatorname{mes} E_n = \sum_{2 \leqslant k \leqslant n-\sqrt{n}} \operatorname{mes} \sigma_k - O\left(\frac{1}{\sqrt{\ln n}}\right) =$$

$$= \left(\frac{4\pi}{2n+1} - \frac{4}{n^2}\right)(n - \sqrt{n}) - O\left(\frac{1}{\sqrt{\ln n}}\right) = 2\pi - o(1),$$

that is,

$$\operatorname{mes} E_n \to 2\pi \quad \text{as} \quad n \to \infty.$$

From (3.32) it follows that

$$|S_{m_k}(x, \varphi_n)| \geqslant \sqrt{\ln n} - 2 = M_n \quad \text{for} \quad x \in E_n^{(k)} \text{ with } 2 \leqslant k \leqslant n - \sqrt{n},$$

where $M_n > 0$, $M_n \to \infty$ as $n \to \infty$. Thus, the conditions (3.2)-(3.4) of Lemma 3.1 are satisfied for $n \geqslant n_0$. If $1 \leqslant n < n_0$, we take $\varphi_n(t) = \varphi_{n_0}(t)$.

This completes the proof of Lemma 3.1.

of detail and difficulty. See, for instance, [7], vol. 1, 430-443; [8], vol. 1, 305-310; [10], [70], [79], Ch. 10.

§4. Further results on divergent Fourier series

Kolmogorov constructed his example in 1922 and published it in 1923 in Fundamenta Mathematicae. On p. 324 he remarks that his methods do not allow him to construct a Fourier series that is everywhere divergent. However, by modifying the construction in §3 somewhat, he constructed in 1926 *an example of an integrable function whose Fourier series diverges everywhere.* (The details are given in [5] or [7], Ch. V.)

These examples were the starting point of much research by other authors. We mention only some of the trends it led to.

1. In 1936 Marcinkiewicz [12] constructed a *function* $f \in L(0, 2\pi)$ *with an almost everywhere divergent Fourier series whose partial sums are bounded at every point of a set* $E \subset [0, 2\pi]$ *with* mes $E = 2\pi$, that is,

$$(4.1) \qquad |s_m(t, f)| \leqslant C(t) \quad \text{for all} \quad \begin{cases} t \in E, \\ m = 0, 1, 2, \ldots \end{cases}$$

We remark (see [20]) that in this theorem one cannot change the function $C(t)$ to a constant C, nor the set E to $[0, 2\pi]$.

For if $|s_m(t, f)| \leqslant C = \text{const}$ for all m and almost all $t \in [0, 2\pi]$, then for all $m \geqslant 0$

$$(4.2) \qquad |\sigma_m(t, f)| = \frac{1}{m+1}\left|\sum_{k=0}^{m} s_k(t, f)\right| \leqslant C$$

for almost all $t \in [0, 2\pi]$.

On the other hand, by Lebesgue's theorem (see [7], vol. 1, 26)

$$(4.3) \qquad \lim_{m\to\infty} \sigma_m(t, f) = f(t) \quad \text{for almost all} \quad t \in [0, 2\pi].$$

From (4.2) and (4.3) it follows that

$$|f(t)| \leqslant C \text{ for almost all } t \in [0, 2\pi],$$

that is, f is essentially bounded.

Next, if (4.1) were to hold for all $t \in [0, 2\pi]$, that is,

$$\overline{\lim_{m\to\infty}} |s_m(t, f)| < \infty,$$

then by Osgood's theorem (see [13], 63) there would be an interval $[a, b]$ and a constant C such that

$$|s_m(t, f)| \leqslant C \quad \text{for all} \quad \begin{cases} t \in [a, b], \\ m = 0, 1, \ldots \end{cases}$$

Arguing as before we would find that $f(t)$ is essentially bounded on $[a, b]$.

Thus, in either case the function $f(t) \in L(0, 2\pi)$ would be essentially bounded on some interval $[a, b]$. Using Riemann's localization principle for Fourier series, we arrive at the conclusion that there are essentially bounded functions whose Fourier series diverge on a set of positive measure, which contradicts Carleson's theorem (see [14] and [15]).

In 1953 Sunouchi ([19]; see also [7], vol. 1, 447–454) remarked that there exists an almost everywhere divergent Fourier series whose conjugate is also a Fourier series.

Now it is standard knowledge that if for some $\varepsilon > 0$ the integral modulus of continuity is

$$(4.4) \qquad \omega_1(\delta, f) = O\left\{\frac{1}{\left(\log\frac{1}{\delta}\right)^{1+\varepsilon}}\right\} \quad \text{as} \quad \delta \to +0,$$

then the Fourier series of f converges almost everywhere on $[0, 2\pi]$. In this connection Zygmund asked in [8], vol. 2 whether this also holds when $\varepsilon = 0$. This is still an open question.

Next, on the basis of Kolmogorov's construction, Prochorenko [21] (see also Tandori [22]) gave an example of a function f with an almost everywhere divergent Fourier series and with the integral modulus of continuity

$$(4.5) \qquad \omega_1(\delta, f) = O\left\{\frac{1}{\log\log\frac{1}{\delta}}\right\} \quad \text{as} \quad \delta \to +0.$$

On the other hand, we have established [23] that

$$\int_0^{2\pi} |f(t)|\,\varphi(|f(t)|)\,dt < \infty$$

if φ is sufficiently smooth and

$$\sum_{n=1}^{\infty} [\varphi(n+1)-\varphi(n)]\,\omega_1\left(\frac{1}{n}, f\right) < \infty. \tag{4.6}$$

If we take $\varphi(t) = (\ln\ln|t|)^{1-\varepsilon}$ for $|t| \geqslant 10$ (ε is any number in $(0, 1)$), we find that by (4.5) and (4.6)

$$\int_0^{2\pi} |f(t)|\,(\ln^+\ln^+|f(t)|)^{1-\varepsilon}\,dt < \infty \quad \text{for all} \quad 0<\varepsilon<1. \tag{4.7}$$

As is not surprising, there are so far no better results that (4.5) and (4.7) on the smoothness and integrability of functions f with divergent Fourier series. By comparing (4.4) and (4.5) we see the extent of our knowledge about the smoothness of functions that have (or do not have) an almost everywhere convergent Fourier series.

2. Another line of research concerns the exact nature of the sets of divergence of Fourier series. There are the standard theorems of Hahn and Sierpinski (see [24], English ed., 305–311) that if we are given a series of continuous functions

$$\sum_{n=1}^{\infty} f_n(t) \qquad (t \in [a, b]), \tag{4.8}$$

[illegible] (or diverges unboundedly) is a $G_{\delta\sigma}$-set (or G_δ-set). The converse is also true, that is, given a set $A \subset [a, b]$ of type $G_{\delta\sigma}$ (or G_δ), we can find a series of continuous functions of the form (4.8) for which the set E of points of divergence (or unbounded divergence) is precisely A.

As regards trigonometric Fourier series, the following result was proved by Zeller [25] on the basis of Kolmogorov's example (see also [20] and [7], Ch. V).

Every set $E \subset [0, 2\pi]$ of type G_δ is a set of unbounded divergence for some Fourier series.

Thus, the class of sets of points of *unbounded* divergence of trigonometric Fourier series of class $L(0, 2\pi)$ is the same as that of G_δ-sets, that is, sets of points of unbounded divergence of series of continuous functions. But this is all we know. For example, if we look at the sets of points of divergence (and not unbounded divergence), they have not been completely characterized either for trigonometric series or for Fourier series of any class (L^p, C, M, and so on). It is a very difficult problem. For instance, Körner [26] has shown that *there exists an F_σ-set $E \subset [0, 2\pi]$* (a fortiori a $G_{\delta\sigma}$-set) *that is not the set of divergence of any trigonometric Fourier series.*

This set E is constructed as follows.

Let $\{t_m\}_{m=1}^{\infty}$ be an everywhere dense set in the half-open interval $[0, 2\pi)$, and $\{E_m\}_{m=1}^{\infty}$ a sequence of closed sets, with $E_m \subset [0, 2\pi)$, each of which is nowhere dense in $[0, 2\pi)$ and such that for any interval I with $x_m \in I$,

$$\operatorname{mes}\{I \cap E_m\} > 0.$$

Then the set

$$E = \bigcup_{m=1}^{\infty} E_m$$

is what we need.

It is quite clear that E is of a very simple structure. However, the proof that E is not the set of divergence of any Fourier series is much more complicated, because it relies (see [26]) on Carleson's theorem (see [14] and [15]). We remark that by construction mes $E > 0$ and in this context Körner [26] notes that the case when mes $E = 0$ remains open.

Finally, we recall a result of Lukashenko [27] to the effect that Körner's set E is not the set of divergence of any trigonometric series, nor of the means of any row-finite Toeplitz summation method for trigonometric series with coefficients that tend to zero. The same kind of result also holds for series in the Walsh system.

3. In §4.1 and §4.2 we have discussed divergence of Fourier series. The question is, what are sufficient conditions for the convergence of Fourier series?

In 1922 Kolmogorov [2] proved the following result.

If $f \in L^2(0, 2\pi)$, then the sequence $s_{n_k}(t, f)$ converges almost everywhere on $[0, 2\pi]$, provided that $n_{k+1} \geqslant \lambda n_k$ for all $k \geqslant 1$ and some $\lambda > 1$.

This was generalized in 1931 by Littlewood and Paley [32] to the case of functions $f \in L^p(0, 2\pi)$ with $p > 1$. Next, Gosselin [33] noted that when $p = 1$ there is no analogous assertion, since for any sequence of natural numbers $n \uparrow \infty$ there exists a function $f \in L(0, 2\pi)$ for which $s_{n_k}(t, f)$ diverges almost everywhere on $[0, 2\pi]$ (and even everywhere on $[0, 2\pi]$, as Totik [78] has established recently).

Next, in [3] Kolmogorov and Seliverstov proved, in particular, that *for any $\epsilon > 0$ the sequence $\{(\log n)^{1+\varepsilon}\}$ is a Weyl multiplier for the convergence almost everywhere of trigonometric series.* Using this method, the same authors in [4] and Plessner in [28] soon showed that the *assertion stated above also holds for $\varepsilon = 0$.*

For forty years this was the best available result on convergence almost everywhere of trigonometric Fourier series of functions $f \in L^2$. Finally, in 1966 Carleson (see [14] and [15]) showed that *the Fourier series of functions $f \in L^2(0, 2\pi)$ converge almost everywhere.* Developing his methods, Hunt [17] showed that the Fourier series of functions in any $L^p(0, 2\pi)$ for $p > 1$ converge almost everywhere. An even stronger assertion

in this direction was proposed by Sjolin [16], who announced that if

$$(4.9)\qquad \int_0^{2\pi} |f(t)|\{\log^+|f(t)|\}\{\log^+\log^+|f(t)|\}\,dt<\infty,$$

then $f(t)$ has an almost everywhere convergent trigonometric Fourier series. Thus, the difference in integrability of functions for which a "positive" and a "negative" result holds (see (4.9) and (4.7)) consists of a factor $\log^+|f(t)|$.

Carleson's proof (see [14] and [15]) is rather complicated, therefore, Fefferman [18] offered a new one (the details are given in [15]). However, Fefferman's proof is also quite complicated, therefore, the search for a simpler proof is of undoubted interest. Furthermore, up to now almost nothing is known on a constructive characterization of those points at which the Fourier series of an arbitrary function $f\in L^2$ necessarily converges almost everywhere on $[0, 2\pi]$ (similarly for functions in C, L^p, and so on).

We remark that [4] has had a great impact, especially in the development of methods of proof to deal with problems on the convergence of orthogonal series. In particular, in 1929 Kaczmarz [29] used this method (see also [9], Ch. V) to show that if for an orthogonal system $\{\varphi_n(t)\}$ the Lebesgue function is

$$L_n(t)\leqslant \omega(n)\uparrow\infty \quad\text{for}\quad t\in E \text{ and } n=1,2,\ldots,$$

then the sequence $\{\omega(n)\}$ is a Weyl multiplier for convergence almost everywhere on E of Fourier series in $\{\varphi_n\}$. Then Tandori [30] showed that this result is best possible in the whole class of ON systems.

In 1936 Marcinkiewicz [12] showed (see also [20], [7], Ch. V, and [11],

If $f\in L(0, 2\pi)$ and on a set $E\subset[0, 2\pi)$ with mes $E>0$

$$\frac{1}{h}\int_0^h |f(t+u)-f(t)|\,du=O\left\{\frac{1}{\ln\frac{1}{|h|}}\right\}\quad \textit{for}\quad t\in E\quad \textit{as}\quad h\to 0,$$

then the Fourier series of $f(t)$ converges almost everywhere on E.

Conversely, *if a non-decreasing even function $\omega(h)$ is such that*

$$\lim_{h\to 0}\omega(h)\ln\frac{1}{h}=\infty,$$

then there exists a function $f\in L(0, 2\pi)$ whose Fourier series diverges almost everywhere on $[0, 2\pi]$ and

$$\frac{1}{h}\int_0^h |f(t+u)-f(t)|\,du=O\{\omega(h)\}\quad \textit{as}\quad h\to 0$$

for almost all $t\in[0, 2\pi]$.

The proof of the second part of this assertion proceeds by using an appropriate modification of Kolmogorov's example in §3.

Finally, we state one of the results of Tsereteli that is related to the problems we have discussed. The following theorem holds (see [31]):

Suppose that $f \in L(0, 2\pi)$. *Then for any* $\varepsilon > 0$ *there exists a rearrangement* ω *of* $(0, 2\pi)$ *such that*

$$\operatorname{mes}\{t\colon\ t \in (0, 2\pi),\ \omega(t) \neq t\} < \varepsilon,$$

and the trigonometric Fourier series of $f(\omega(t))$ *converges almost everywhere.*

4. In [1] Kolmogorov constructed an integrable function $\Phi(t)$ whose trigonometric Fourier series diverges almost everywhere. In this paper he says ([1], 324) that he did not know the order of magnitude of the Fourier coefficients of his function Φ. As far as we know, this question has still not been investigated in detail.

In the same connection there arises another problem.

Suppose that a sequence $\{\varepsilon_n\} \notin l_2$ and $\varepsilon_n \downarrow 0$. For what sets $E \subset [0, 2\pi)$ can we construct a function $F \in L(0, 2\pi)$ whose trigonometric Fourier series on $[0, 2\pi)$ has E as its set of divergence and

$$\sqrt{a_n^2(F) + b_n^2(F)} \leqslant \varepsilon_n \quad \text{for} \quad n \geqslant 0. \tag{4.10}$$

Of course, we can look at other classes of functions (not just L) and other systems of functions (not just the trigonometric system).

A recent result along these lines is due to Galstyan [71]:

For any sequence $\{\varepsilon_n\} \notin l_2$ *and* $\varepsilon_n \downarrow 0$ *there exists a function* $F \in L(0, 2\pi)$ *whose trigonometric Fourier series diverges everywhere, moreover,* (4.10) *holds.*

The proof of this proposition is based on Kolmogorov's method [5], who in 1926 constructed a Fourier series that diverges everywhere (see also [8], vol. I, 310–314).

5. We now mention some results that refer to other systems of functions.

In 1960 Stein [70] proved that *there exists a function* $f \in L(0, 1)$ *such that its Fourier-Walsh-Paley series diverges a.e. on* $[0, 1]$. This result was sharpened by Schipp [82], who proved an analogue of Prochorenko's theorem for the Walsh system (that is, the function f in Stein's theorem can be chosen to satisfy a condition of the form (4.5)).

Next, in 1961 (see [38], 72–73, and [45], 15) we drew attention to the importance of obtaining a lower bound for the Lebesgue functions of orthonormal systems $\{\varphi_n\}$ on $[0, 1]$ with

$$|\varphi_n(t)| \leqslant M = \text{const} \qquad \text{for all} \quad \begin{cases} n = 1, 2, \ldots, \\ t \in [0, 1]. \end{cases} \tag{4.11}$$

The first very general result in this direction was obtained by Olevskii in 1965–1966 (see [43], [44], and [46]). The most general result (see [44] and [46]) is that

$$\max_{1 \leqslant m \leqslant N} \int_0^1 \Big| \sum_{n=1}^{m} a_n \varphi_n(t) \Big| dt \geqslant C(M) \frac{\log N}{N} \sum_{n=1}^{N} a_n^2 \tag{4.12}$$

for any ONS satisfying (4.11), for any $\{a_n\}$ with $|a_n| \leqslant M$ for $n \geqslant 1$ and all $N = 1, 2, \ldots$ Here the constant $C(M) > 0$ depends only on M.

If we set $a_n = \varphi_n(y)$ in (4.12), we find that the Lebesgue functions $L_N(y)$ for the system $\{\varphi_n\}$ satisfy the relation (see [44])

$$(4.13) \qquad \overline{\lim_{N\to\infty}} \frac{L_N(y)}{\log N} > 0 \quad \text{for} \quad y \in E \subset [0, 1] \quad \text{with } mE > 0.$$

It is clear that (4.13) is the best possible estimate, since even for the trigonometric system the Lebesgue functions (Lebesgue constants) are asymptotically equal to $\log N$.

Using these results Olevskii [44] deduced that for any ONS $\{\varphi_n\}$ that is uniformly bounded there is a continuous function whose Fourier series in $\{\varphi_n\}$ diverges at some point y_0, and the rate of divergence can be made arbitrarily close to $\log N$ (this is a sharpened analogue of the theorem of du Bois-Reymond). In addition, for every positive sequence $\{\omega(n)\}$ with $\omega(n) = o(\log n)$ as $n \to \infty$ there exists a function $f \in L(0, 1)$ such that

$$\overline{\lim_{N\to\infty}} \frac{1}{\omega(N)} \| S_N(t, f) \|_L = \infty,$$

where $S_N(t, f)$ are the partial sums of the Fourier series of f in the system $\{\varphi_n\}$.

Next, Alexits (see [11], English ed., 326, and [84], 287) and Olevskii (see [80], 1428) posed the problem of an analogue of Kolmogorov's example (theorem) of a divergent Fourier series for ON systems that are uniformly bounded.

A partial answer to this question[1] was given by Bochkarev [57] in 1975: *given any ONS* $\{\varphi_n\}$ *satisfying* (4.11), *there exists a function* $f \in L(0, 1)$ *whose Fourier series in* $\{\varphi_n\}$ *diverges at every point of some set* E *of*

At the basis of the proof there lies, on the one hand, Kolmogorov's idea (see [57], Lemma 4) of the construction of a discrete measure whose Fourier-Stieltjes series has the greatest majorants of the partial sums on sets of large measure (this idea had been used repeatedly before; see, for example, [70], [79], Ch. 10, and [83]), and on the other hand it also uses the inequality ([57], Lemma 3)

$$(4.14) \qquad \frac{1}{N} \sum_{m=1}^{N} \int_0^1 \Big| \sum_{n=1}^{m} a_n \varphi_n(t) \Big| dt + \int_0^1 \Big| \sum_{n=1}^{N} a_n \varphi_n(t) \Big| dt \geqslant C(M) \frac{\log N}{N} \sum_{n=1}^{N} a_n^2.$$

To all appearances, the proof of the analogue of Kolmogorov's theorem in Bochkarev's paper [57] can be shortened. Of the four lemmas in [57] the first three are concerned with establishing the final inequality (4.14). However, in the proof of the analogue of Kolmogorov's theorem in [57]

[1] Probably the author of [57] did not know who had raised the question and he did not give the relevant references.

only the old inequality (4.12) of Olevskii is actually used, therefore, we have not understood the purpose of the first three lemmas (nor of (4.14)).[1]

A related result is the assertion that there cannot be, in general, a complete analogue of Kolmogorov's example for ON systems. This follows from the fact that Kazaryan [58] has constructed *a complete ON system that is uniformly bounded and for which every Fourier series converges on some set of positive measure.*

§5. Kolmogorov's theorem on the divergence of Fourier series of class L^2 in a rearranged trigonometric system, and some generalizations of it

In §1 we spoke of Kolmogorov's paper [6] in which he announced the following theorem.

***Theorem* 5.1.** *There exists an almost everywhere divergent series of the form*

$$\sum_{n=1}^{\infty} a_n \cos(m_n t + \lambda_n),$$

where the integers m_n $(n = 1, 2, \ldots)$ *are all distinct and the series*

$$\sum_{n=1}^{\infty} a_n^2$$

converges.

In other words, the trigonometric system is not a system of unconditional convergence almost everywhere.

This theorem was stated in 1926, but was not proved until 1960, when Zagorskii [34] gave a short sketch of a proof. At the same time in connection with Kolmogorov's theorem we posed the question (see [35], 452, Problem 3) whether there exist complete orthonormal systems of unconditional convergence almost everywhere. We proved that if there are such systems, then one of them must be the Haar system ([35], 452). However, by modifying Zagorskii's construction suitably we showed ([36], [38]) that the Haar system does not have this property.

This result immediately suggests that probably there are no complete orthonormal systems of unconditional convergence almost everywhere at all (see [45], 48). Indeed, on the basis of the above assertion about the Haar system, the author ([37] and [38]) and Olevskii [41] have obtained corroborative results in this direction. Moreover, in [37] and [38] we showed that L^2 has no basis that is a system of unconditional convergence almost everywhere.

Before we give more recent results on this, we give now the proof of the assertions just stated, following [38] and [40].

[1]We refer the reader to Kashin's paper [85], in which, in particular, there is a simple proof that the Lebesgue functions are bounded below.

***Lemma* 5.1** (see [40]). *Suppose that* $f_n(t) \in L(0, 1)$ $(n = 1, 2, \ldots, m)$ *and*

$$Q(t) = \sum_{n=1}^{\infty} a_n f_n(t).$$

Then if

$$Q^*(t) = \sup_{1 \leqslant i \leqslant m} \Big| \sum_{n=1}^{i} a_n f_n(t) \Big|,$$

we have

$$\int_0^1 Q^*(t)\, dt \geqslant \sum_{n=1}^{m} a_n \int_{E_n} f_n(t)\, dt \tag{5.1}$$

for any sequence of measurable sets

$$E_1 \supset E_2 \supset \ldots \supset E_m.$$

Proof. Clearly,

$$\sum_{n=1}^{i} a_n f_n(t) \leqslant Q^*(t) \quad \text{with} \quad t \in [0, 1] \quad \text{and} \quad 1 \leqslant i \leqslant m. \tag{5.2}$$

Therefore, putting $E_{m+1} = 0$ we obtain (see (5.2))

$$\sum_{n=1}^{m} a_n \int_{E_n} f_n(t)\, dt = \int_{E_m} (a_1 f_1 + a_2 f_2 + \ldots + a_m f_m)\, dt +$$

$$+ \int_{E_{m-1} \setminus E_m} (a_1 f_1 + \ldots + a_{m-1} f_{m-1})\, dt + \ldots$$

$$\ldots + \int_{E_n \setminus E_{n+1}} (a_1 f_1 + \ldots + a_n f_n)\, dt + \ldots + \int_{E_1 \setminus E_2} a_1 f_1\, dt =$$

$$= \sum_{n=1} \int_{E_n \setminus E_{n+1}} (a_1 f_1 + \ldots + a_n f_n)\, dt \leqslant \sum_{n=1} \int_{E_n \setminus E_{n+1}} Q^*(t)\, dt \leqslant$$

$$\leqslant \int_{E_1} Q^*(t)\, dt \leqslant \int_0^1 Q^*(t)\, dt,$$

that is, (5.1) holds, and Lemma 5.1 is proved.

We now prove the following result ([37]-[40]).

***Theorem* 5.2.** *There exists a series*

$$\sum_{m=1}^{\infty} a_m \chi_m(t) \quad \textit{with} \quad \sum_{m=1}^{\infty} a_m^2 < \infty, \tag{5.3}$$

which after an appropriate rearrangement diverges boundedly almost everywhere on $[0, 1]$.

Proof. Let N be any natural number. We split $[0, 1]$ into 2^N equal parts of length $1/2^N$, and we set $t_i = (i-1)/2^N$ and $E_i = (t_i, 1]$, where $i = 2, 3, \ldots, 2^N$. Clearly, the E_i decrease as i increases.

We consider the Haar functions

$$\chi_n^{(k)} \text{ with } 0 \leqslant n \leqslant N-1 \text{ and } 1 \leqslant k \leqslant 2^n, \text{ i.e. } \chi_m(t) \text{ with } 2 \leqslant m \leqslant 2^N,$$

and arrange (renumber) them so that in the new order $\chi_{m_2}(t), \chi_{m_3}(t), \ldots, \chi_{m_{2^N}}(t)$ the integrals

$$\left|\int_{E_i} \chi_{m_i}(t)\,dt\right| \quad (2 \leqslant i \leqslant 2^N)$$

are maximal for each integer $i \in [2, 2^N]$. This order is equivalent to the requirement that the essential zeros of the functions χ_{m_i} come in the order of increasing indices i $(2 \leqslant i \leqslant 2^N)$. The $\chi_{m_i}(t)$ will have essential zeros at the points t_i $(2 \leqslant i \leqslant 2^N)$. In this case we find that

$$\int_{E_i} \chi_{m_i}(t)\,dt = -\frac{1}{2}\int_0^1 |\chi_{m_i}(t)|\,dt. \tag{5.4}$$

We now take the polynomial

$$Q_N(t) = \sum_{i=2}^{2^N} q_i \chi_{m_i}(t) \quad \text{with} \quad q_i = \frac{2}{N}\int_{E_i} \chi_{m_i}(x)\,dx, \tag{5.5}$$

which plays the main role in the subsequent constructions.

Here are some of its properties.

Firstly, if the terms of $Q_N(t)$ are in their natural order, then (by (5.4), (5.5), and §2) we find that

$$Q_N(t) = -\frac{1}{N}\sum_{n=0}^{N-1}\sum_{k=1}^{2^n}\left(\int_0^1 |\chi_n^{(k)}(x)|\,dx\right)\chi_n^{(k)}(t) = -\frac{1}{N}\sum_{n=0}^{N-1} r_n(t). \tag{5.6}$$

Thus, (see (5.6))

$$\int_0^1 Q_N^2(t)\,dt = \frac{1}{N^2}\int_0^1 \left\{\sum_{n=0}^{N-1} r_n(t)\right\}^2 dt = \frac{1}{N}. \tag{5.7}$$

Secondly, by Lemma 5.1, (see (5.1), (5.5), and (5.4)),

$$\int_0^1 Q_N^*(t)\,dt \geqslant \sum_{i=2}^{2^N} q_i \int_{E_i} \chi_{m_i}(t)\,dt = \frac{2}{N}\sum_{i=2}^{2^N}\left\{\int_{E_i} \chi_{m_i}(x)\,dx\right\}^2 = \tag{5.8}$$

$$= \frac{1}{2N}\sum_{n=0}^{N-1}\sum_{k=1}^{2^n}\left\{\int_0^1 |\chi_n^{(k)}(x)|\,dx\right\}^2 = \frac{1}{2N}\sum_{n=0}^{N-1} 1 = \frac{1}{2}.$$

Thirdly, at each fixed irrational point $t_0 \in [0, 1]$ only one function from the group $\chi_n^{(k)}(t_0)$ (n fixed and $1 \leqslant k \leqslant 2^N$) is non-zero. Thus, by (5.5) and (5.4)

$$Q_N^*(t) \leqslant \frac{2}{N}\sum_{i=2}^{2^N}\left|\int_{E_i} \chi_{m_i}(x)\,dx\right| \cdot |\chi_{m_i}(t)| = \tag{5.9}$$

$$= \frac{1}{N}\sum_{n=0}^{N-1}\sum_{k=1}^{2^n}\left\{\int_0^1 |\chi_n^{(k)}(t)|\,dx\right\}|\chi_n^{(k)}(t)| = \frac{1}{N}\sum_{n=0}^{N-1} 1 = 1$$

for almost all $t \in [0, 1]$.

From the definition (5.5) of $Q_N(t)$ it is clear that it is a step-function with constant step length $1/2^N$. This implies that $Q_N^*(t)$ is also a step-function with constant step length $1/2^N$.

Finally, if we set

$$A=\left\{t\colon\ t\in[0,\ 1],\quad Q^*(t)\geqslant\frac{1}{4}\right\},\qquad B=[0,\ 1]\setminus A,$$

we obtain (see (5.8) and (5.9))

$$\frac{1}{2}\leqslant\int_0^1 Q_N^*(t)\,dt=\int_A Q_N^*(t)\,dt+\int_B Q_N^*(t)\,dt\leqslant mA+\frac{1}{4}\,mB\leqslant mA+\frac{1}{4},$$

that is,

$$mA=\left|\left\{t\colon\ t\in[0,\ 1],\quad Q_N^*(t)\geqslant\frac{1}{4}\right\}\right|\geqslant\frac{1}{4}. \tag{5.10}$$

The properties (5.7)–(5.10) of $Q_N(t)$ enable us to construct a function $F(t)\in L^2(0,\ 1)$ whose Fourier series satisfies the requirements of Theorem 5.2.

We now turn to the direct construction of $F(t)$. Let $\{N_i\}_{i=1}^{\infty}$ be an arbitrary increasing sequence of natural numbers such that

$$\sum_{i=1}^{\infty}\frac{1}{N_i}<\infty. \tag{5.11}$$

We put $P_1(t)=Q_{N_1}(t)$. Clearly, by (5.10),

$$mE\left\{t\colon\ t\in[0,\ 1],\quad P_1^*(t)\geqslant\frac{1}{4}\right\}\geqslant\frac{1}{4}.$$

Now we put

$$P_2(t)=Q_{N_2}(2^{-1}t). \tag{5.12}$$

This is the second step of the construction. We recall that the polynomial $Q_{N_2}(T)$ consists of the Haar functions $\chi_n^{(k)}(t)$ with $0\leqslant n\leqslant N_2-1$ and $1\leqslant k\leqslant 2^n$.

But for any integers $j\geqslant 1$

$$\chi_n^{(k)}(2^jt)=\frac{1}{\sqrt{2^j}}\{\chi_{n+j}^{(k)}(t)+\chi_{n+j}^{(1\cdot 2^n+k)}(t)+ \tag{5.13}$$
$$+\chi_{n+j}^{(2\cdot 2^n+k)}(t)+\ldots+\chi_{n+j}^{((2^j-1)2^n+k)}(t)\},$$

which follows directly from the definition of the Haar functions. Therefore, $P_2(t)$ consists of the Haar functions (see (5.12) and (5.13)) with the indices

$$\chi_n^{(k)}(t)\ \text{ with }\ N_1\leqslant n\leqslant N_1+N_2-1\ \text{ and }\ 1\leqslant k\leqslant 2^n.$$

Hence, $P_2(t)$ does not have terms in common with $P_1(t)$, which consists of the Haar functions

$$\chi_n^{(k)}(t)\ \text{ with }\ 0\leqslant n\leqslant N_1-1\ \text{ and }\ 1\leqslant k\leqslant 2^n.$$

Next, we know (see (5.10)) that

$$mE\left\{t\colon\ t\in[0,\ 1],\quad Q_{N_2}^*(t)\geqslant\frac{1}{4}\right\}\geqslant\frac{1}{4}. \tag{5.14}$$

Let $B_2 = E\left\{t\colon t \in [0, 1],\ P_2^*(t) \geqslant \frac{1}{4}\right\}$. Then by (5.12) and (5.14)

$$m\left\{B_2\left(\frac{i-1}{2^{N_1}}, \frac{i}{2^{N_1}}\right)\right\} \geqslant \frac{1}{4}\frac{1}{2^{N_1}} \quad \text{for all} \quad 1 \leqslant i \leqslant 2^{N_1}.$$

Suppose that we have already constructed the polynomials

$$P_1(t), \ldots, P_{s-1}(t)$$

so that they have no terms in common and that $P_{s-1}(t)$ consists of the Haar functions

$$\chi_n^{(k)}(t) \text{ with } N_1 + \ldots + N_{s-2} \leqslant n \leqslant N_1 + \ldots + N_{s-1} - 1 \text{ and } 1 \leqslant k \leqslant 2^n.$$

We take the polynomial

$$P_s(t) = Q_{N_s}(2^{N_1+\ldots+N_{s-1}}t), \tag{5.15}$$

which consists of the Haar functions (see (5.6) and (5.13))

$$\chi_n^{(k)}(t) \text{ with } N_1 + \ldots + N_{s-1} \leqslant n \leqslant N_1 + \ldots + N_s - 1 \text{ and } 1 \leqslant k \leqslant 2^n$$

and, therefore, has no terms in common with $P_{s-1}(t)$. We note that $P_s(t)$ has least intervals of constant length $1/2^{N_1+\ldots+N_s}$ (see (5.15)). We set

$$B_s = E\left\{t\colon t \in [0, 1],\ P_s^*(t) \geqslant \frac{1}{4}\right\}.$$

Then just as we have done at the second step, we find by (5.15) and (5.10) that

$$mE\left\{B_s\left(\frac{i-1}{2^{N_1+\ldots+N_{s-1}}}, \frac{i}{2^{N_1+\ldots+N_{s-1}}}\right)\right\} \geqslant \frac{1}{4}\frac{1}{2^{N_1+\ldots+N_{s-1}}} \tag{5.16}$$

for all $1 \leqslant i \leqslant 2^{N_1+\ldots+N_{s-1}}$.

From (5.15) and (5.17) we have

$$\int_0^1 P_s^2(t)\,dt = \frac{1}{N_s} \quad \text{for} \quad s = 1, 2, \ldots \tag{5.17}$$

We consider the series

$$\sum_{s=1}^{\infty} P_s(t), \tag{5.18}$$

which is the Fourier series of some function $F(t) \in L^2(0, 1)$, since the $P_s(t)$ are mutually orthogonal and (5.11) holds (see also (5.17)).

Now the partial sums of (5.18):

$$A_q(t) = \sum_{s=1}^{q} P_s(t) \quad (q = 1, 2, \ldots) \tag{5.19}$$

converge almost everywhere on $[0, 1]$, since they form a subsequence of the partial sums of the Fourier-Haar series of $F(t)$. In fact, any subscript n in the Haar functions $\chi_n^{(k)}(t)$ of $P_s(t)$ is strictly less than any subscript in the Haar functions in $P_{s+1}(t)$. This implies (see (5.19)) that

$$A_q(t) = S_{2^{N_1+\ldots+N_q}}(t, F) \quad (q = 1, 2, \ldots). \tag{5.20}$$

Suppose now that in (5.18) the polynomial $P_s(t)$ is written in the form (5.5), where N is replaced by N_s and t by $2^{N_1+\dots+N_{s-1}}t$, and that (5.13) is applicable. Then we obtain a certain series in the rearranged Haar system, which is a rearrangement of the Fourier-Haar series of $F(t)$.

We now verify that the rearranged series (5.18) diverges almost everywhere on $[0, 1]$. For let

$$B=\overline{\lim_{s\to\infty}}\, B_s, \text{ that is, } B=\bigcap_{k=2}^{\infty}\bigcup_{s=k}^{\infty} B_s.$$

Clearly, if $t_0 \in B$ then $t_0 \in B_s$ for infinitely many s, therefore, $P_s^*(t_0) \geqslant 1/4$ for infinitely many s. This implies that the rearranged series (5.18) diverges at t_0, since it does not satisfy the Cauchy criterion. Thus, we need only check that $mB = 1$. But from (5.16) it follows that

$$m\{([0, 1]\setminus B_{s-1})\}\leqslant\frac{3}{4}, \quad m\{([0, 1]\setminus B_{s-1})([0, 1]\setminus B_s)\}\leqslant\left(\frac{3}{4}\right)^2,$$

and, more generally,

$$m\{([0, 1]\setminus B_{s-1})\cap([0, 1]\setminus B_s)\cap\dots\cap([0, 1]\setminus B_{s+p})\}\leqslant\left(\frac{3}{4}\right)^{p+2} \text{ for } p\geqslant 0.$$

Therefore,

$$m\bigcap_{i=s-1}^{\infty}([0, 1]\setminus B_i)=\lim_{p\to\infty}\left(\frac{3}{4}\right)^{p+2}=0 \quad (s=3, 4, \dots).$$

Hence,

$$m\{[0, 1]\setminus B\}=m\bigcup_{k=2}^{\infty}\bigcap_{s=k}^{\infty}([0, 1]\setminus B_s)=0, \text{ that is, } mB=1.$$

On the other hand, since $A_q(t)$ converges almost everywhere (see (5.20)) it [illegible]

almost everywhere on $[0, 1]$. Moreover, the amplitude of the oscillations cannot exceed 2. This completes the proof of Theorem 5.2.

Remark 5.1. There are now some simpler methods of constructing examples of divergent rearrangements of Fourier-Haar series of class L^2 (see, for example, the proof of Theorem 3 in [40], or [47]). However, the proof given here can be instructive and is useful for subsequent investigations of applications of the methods.

Remark 5.2. Our example of a function $F \in L^2(0, 1)$ whose rearranged Fourier-Haar series diverges *boundedly* almost everywhere on $[0, 1]$ is the first of its kind. We do not know of any such examples for other classical systems of functions nor, a fortiori, for fairly arbitrary complete ON systems.

Remark 5.3. From the proof of Theorem 5.2 it is clear that it can be supplemented in various directions. For instance, the following assertions are true (see [37] and [38]).

***Theorem* 5.3.** *There exists a series of the form* (5.3) *which after a suitable rearrangement of its terms diverges unboundedly almost everywhere on* $[0, 1]$.

***Theorem* 5.4.** *For any sequence of natural numbers $n_m \uparrow \infty$ there exists a series of class L^2:*

$$\sum_{m=1}^{\infty} \{\sum_{k=1}^{2^{n_m}} a_{n_m}^{(k)} \chi_{n_m}^{(k)}(t)\} = \sum_{m=1}^{\infty} A_m(t),$$

which after a suitable rearrangement of the $A_m(t)$ diverges to infinity almost everywhere on $[0, 1]$.

Theorem 5.3 follows directly from Theorem 5.2 and the following simple fact:

If $\sum a_n$ diverges boundedly, then any series $\sum \lambda_n a_n$ diverges to infinity, provided that $\lambda_n \uparrow \infty$.

The proof of Theorem 5.4 proceeds by analogy with that of Theorem 5.2 (some indication is given in [40]).

Now we can verify the following result (see [37] and [38]).

***Theorem* 5.5.** *For any basis $\{f_i\}_{i=1}^{\infty}$ in $L^2(0, 1)$ there exists a function $F \in L^2(0, 1)$ such that its Fourier series*[1]

$$F(t) = \sum_{k=1}^{\infty} \alpha_k f_k(t) \qquad (\alpha_k = (F, \psi_k)) \tag{5.21}$$

after some rearrangement of its terms diverges to infinity almost everywhere on $[0, 1]$. *Here $\{\psi_k\}$ is the conjugate system to $\{f_k\}$.*

The proof is based on Theorem 5.4, Marcinkiewicz's method (see [49] and [9]) and Banach's theorem on the existence of a conjugate system for any basis (see [48] and [9]).

Since $\{f_i\}$ is a basis in L^2 and $\{\chi_m\}$ is a Haar system,

$$\chi_i(t) \overset{L^2}{=} \sum_{\nu=1}^{\infty} a_\nu^{(i)} f_\nu(t), \text{ where } a_\nu^{(i)} = (\chi_i, \psi_\nu) = \int_0^1 \chi_i \psi_\nu \, dt \quad (i = 1, 2, \ldots) \tag{5.22}$$

and by Parseval's identity for the Haar system

$$\sum_{i=1}^{\infty} (a_\nu^{(i)})^2 = \int_0^1 \psi_\nu^2 \, dt \qquad (\nu = 1, 2, \ldots). \tag{5.23}$$

Thus, (see (5.22) and (5.23)) we can find strictly increasing sequences of natural numbers $\{q_i\}$ and $\{p_i\}$ such that

$$q_i \geqslant i, \quad \sum_{i=p_\nu}^{\infty} (a_\nu^{(i)})^2 < \frac{1}{4^\nu} \quad (\nu \geqslant 1) \tag{5.24}$$

$$\text{and} \quad \int_0^1 \Big| \chi_i - \sum_{\nu=1}^{q_i} a_\nu^{(i)} f_\nu \Big|^2 dt < \frac{1}{2^i} \qquad (i \geqslant 1).$$

We set

$$u(i) = \begin{cases} 1 & \text{for} \quad 1 \leqslant i < p_1, \\ k & \text{for} \quad p_k \leqslant i < p_{k+1}. \end{cases} \tag{5.25}$$

[1](5.21) is understood in the sense of convergence of the series to F in the L^2-norm.

Clearly, $q_i > i \geqslant u(i)$ and $u(i) \uparrow \infty$ (see (5.24) and (5.25)). Now we find increasing sequences of natural numbers $\{m_k\}$ and $\{n_k\}$ such that

$$(5.26) \qquad m_1 = 1, \quad m_{2k} < 2^{n_k} < 2^{n_k+1} < m_{2k+1} \quad \text{and } u(m_{k+1}) > q_{m_k} \quad \text{for} \quad k \geqslant 1.$$

In accordance with (5.22) we can represent χ_i in the form

$$(5.27) \qquad \chi_i = \sum_{\nu=1}^{u(i)} a_\nu^{(i)} f_\nu + \sum_{\nu=u(i)+1}^{q_i} a_\nu^{(i)} f_\nu + [\chi_i - \sum_{\nu=1}^{q_i} a_\nu^{(i)} f_\nu] = \varphi_i' + \varphi_i'' + \varphi_i''' \qquad (i \geqslant 1).$$

We consider the sequence $\{b_i\} \in l_2$ with $\sum_{i=1}^{\infty} b_i^2 = D$. Formally,

$$(5.28) \qquad \sum_{i=1}^{\infty} b_i \chi_i = \sum_{i=1}^{\infty} b_i \varphi_i' + \sum_{i=1}^{\infty} b_i \varphi_i'' + \sum_{i=1}^{\infty} b_i \varphi_i''' = R_1 + R_2 + R_3,$$

where the series on the left converges in the L_2-norm.

For the series R_3 we have (see (5.24))

$$\int_0^1 \Big\{ \sum_{i=1}^{\infty} |b_i \varphi_i'''(t)| \Big\}^2 dt \leqslant \sum_{i=1}^{\infty} b_i^2 \sum_{i=1}^{\infty} \int_0^1 \{\varphi_i'''(t)\}^2 dt \leqslant \sum_{i=1}^{\infty} b_i^2 \sum_{i=1}^{\infty} \frac{1}{2^i} = D,$$

that is, R_3 is absolutely convergent almost everywhere in $[0, 1]$. As for R_1, we have

$$\sum_{i=1}^{\infty} |b_i \varphi_i'| \leqslant \sum_{i=1}^{\infty} |b_i| \sum_{1 \leqslant \nu \leqslant u(i)} |a_\nu^{(i)}| \cdot |f_\nu| \leqslant$$

$$\leqslant \sum_{\nu=1} \Big\{ \sum_{\substack{i \\ u(i) \geqslant \nu}} |b_i| \cdot |a_\nu^{(i)}| \Big\} |f_\nu| = \sum_{\nu=1} a_\nu |f_\nu(t)|,$$

where (see (5.25) and (5.24))

$$0 \leqslant d_\nu \leqslant \sum_{i=p_\nu}^{\infty} |b_i| \cdot |a_\nu^{(i)}| \leqslant \Big\{ \sum_{i=p_\nu}^{\infty} b_i^2 \Big\}^{1/2} \Big\{ \sum_{i=p_\nu}^{\infty} |a_\nu^{(i)}|^2 \Big\}^{1/2} \leqslant \frac{\sqrt{D}}{2^\nu}.$$

Therefore,

$$\sum_{i=1}^{\infty} \int_0^1 |b_i \varphi_i'| \, dt \leqslant \sum_{\nu=1}^{\infty} \frac{\sqrt{D}}{2^\nu} \int_0^1 |f_\nu(t)| \, dt \leqslant \sum_{\nu=1}^{\infty} \frac{\sqrt{D}}{2^\nu} = \sqrt{D},$$

which implies the absolute convergence almost everywhere of R_1.

In roughly the same way we can prove that R_1 and R_3 converge in $L^2(0, 1)$. But then it follows from (5.28) (see also (5.27)) that

$$(5.29) \qquad R_2 = \sum_{i=1}^{\infty} b_i \varphi_i'' = \sum_{i=1}^{\infty} b_i \Big\{ \sum_{\nu=u(i)+1}^{q_i} a_\nu^{(i)} f_\nu(t) \Big\}$$

also converges in the norm of $L^2(0, 1)$.

We set

$$b_i = \begin{cases} c_i & \text{for } 2^{n_k}+1 \leqslant i \leqslant 2^{n_k+1} \quad (k=1,\ 2,\ \ldots), \\ 0 & \text{for the remaining } i, \end{cases}$$

assuming that $\{b_i\} \in l_2$. Let

$$B_k(t) = \sum_{i=2^{n_k}+1}^{2^{n_k+1}} c_i \Big\{ \sum_{\nu=u(i)+1}^{q_i} a_\nu^{(i)} f_\nu(t) \Big\}.$$

Then (see (5.26))

$$(5.30) \qquad B_k(t) = \sum_{j=u(m_{2k})}^{q_{m_{2k+1}}} l_j f_j(t) \qquad (l_j \text{ are certain constants})$$

therefore, the polynomials B_k do not have common terms. Since (5.29) converges in L^2, from

$$(5.31) \qquad \sum_{k=1}^{\infty} \Big\{ \sum_{i=2^{n_k}+1}^{2^{n_k+1}} c_i \chi_i(t) \Big\} = R_1 + R_3 + \sum_{k=1}^{\infty} B_k(t)$$

it follows that the series

$$(5.32) \qquad \sum_{k=1}^{\infty} B_k(t)$$

converges in L^2 to some function $F(t) \in L^2(0,\ 1)$. This implies that if we substitute the terms B_k from (5.30) in (5.32), we obtain a series

$$(5.33) \qquad \sum_{j=1}^{\infty} l_j f_j,$$

that has some subsequence of partial sums converging in L^2 to F. By Lemma 2.2, this means that (5.33) is the Fourier series of F in the basis $\{f_j\}_{j=1}^{\infty}$.

We know that by Theorem 5.4 we can choose numbers c_n so that after some rearrangement of the $A_k(t)$ the series

$$\sum_{k=1}^{\infty} A_k(t) = \sum_{k=1}^{\infty} \Big\{ \sum_{i=2^{n_k}+1}^{2^{n_k+1}} c_i \chi_i(t) \Big\}$$

diverges to infinity almost everywhere on $[0,\ 1]$. But R_1 and R_3 always converge absolutely almost everywhere, therefore it follows from (5.31) that for this choice of the c_i the functions $B_k(t)$ in (5.32) can be rearranged to give a series that diverges to infinity almost everywhere. But every rearrangement of (5.32) is also a rearrangement of (5.33). This completes the proof.

If we use one of our results in the theory of summability ([38], §3), then as an immediate consequence of Theorem 5.5 we obtain the following result.

***Theorem* 5.6.** *For every basis $\{f_i\}$ in L^2 there is a function $F \in L^2$ such that for any Toeplitz summability method T the terms of* (5.21) *can be rearranged so that the resulting series is not T-summable at almost every point.*

These results gave a fresh impetus to the subsequent work of numerous authors (Ul'yanov, Olevskii, A.A. Talalyan, Tandori, Arutyunyan, Tsereteli, Taikov, Moritz, F.A. Talalyan, Leindler, Nikishin, Nakata, Bochkarev, Kashin, Poleshchuk, and others). Here we mention only a few of their results.

In [38] we established that there exist Fourier series of class L^p (for any $p > 2$) in the Haar system (or any complete uniformly bounded ONS) that do not converge unconditionally. This result was sharpened by Olevskii [42], who proved the following theorem.

***Theorem* 5.7.** *For any complete ONS $\{\varphi_m\}$ there is a continuous function whose Fourier series in the system $\{\varphi_m\}$ diverges to infinity almost everywhere after a suitable rearrangement of its terms.*

Next, A.A. Talalyan [60] has shown that Theorem 5.5 can be generalized to arbitrary systems $\{f_m\}$ in which functions of class $L_q(0, 1)$ $(q > 1)$ have a representation by a series that converges in the norm of $L(0, 1)$.

It is not clear what is the minimal class of functions that can be taken in place of L_q for this theorem to remain valid. This problem is, in fact, raised in [61] (see p.55).

Tsereteli has proved another type of result [59].

***Theorem* 5.8.** *If $\{\varphi_m\}$ is a complete ON system and $f \in L^2(0, 1)$, then there is a rearrangement π of the natural numbers such that for any $\varepsilon > 0$ we can find an f_ε with*

$$|f_\varepsilon(t)| = |f(t)| \quad \text{for } t \in [0, 1] \text{ and } m\{t: f_\varepsilon(t) \neq f(t)\} < \varepsilon,$$

whose Fourier series in the system $\{\varphi_{\pi(m)}\}$ diverges to infinity almost everywhere on $N(f) = E\{t: f(t) \neq 0\}$.

In 1961 we established the following result (see [39] and [40]).

***Theorem* 5.9.** *A positive sequence $\omega(m)\uparrow$ is a Weyl multiplier for unconditional convergence almost everywhere of series in the Haar system $\{\chi_m\}$, that is, $\{\chi_m\} \in W(u\,c, \omega(m))$ if and only if*

$$\sum_{m=1}^{\infty} \frac{1}{m\omega(m)} < \infty. \tag{5.34}$$

As far as we know, this is the first result on classical ONS for which the question of Weyl multipliers for unconditional convergence almost everywhere is completely solved.

We mention some results connected with Theorem 5.9.

1) It follows from Theorem 5.9 that the Haar system does not have an exact Weyl multiplier for unconditional convergence almost everywhere.

2) The inequality (5.34), which expresses a necessary and sufficient condition for $\{\omega(m)\}$ to be a Weyl multiplier, probably first appeared in [39].

It has enabled us to solve definitively the problem of Weyl multipliers for absolute $|c, \alpha|$-summability of orthogonal series. For example, we have shown (see [45], 38–46) that for every class of orthogonal series (and even only for the class of trigonometric series) the sequence $\{\omega(m)\}$ is a Weyl multiplier for $|c, \alpha|$-summability almost everywhere ($\alpha > 1/2$ is arbitrary but fixed) if and only if (5.34) is satisfied.

3) Propositions like Theorem 5.9 have naturally led us to raise (and almost solve) the problem of finding optimal Weyl multipliers for the trigonometric system and the Walsh system (see [37]–[40]). Moreover, we have even suggested (see [45], 50) that the growth of the best possible Weyl multiplier for these systems is roughly of order $\log m$ or $\log^2 m$. Of course, we can also look at other specific systems (Legendre polynomials, orthogonal polynomials, the Franklin system, and others).

What is known today?

The first move towards a partial solution of the problem for the trigonometric system was made by Moritz [53] in 1966, who showed that

$$\{\cos mt,\ \sin mt\} \notin W(uc,\ [0,\ 2\pi],\ \omega(m)), \tag{5.35}$$

if $\omega(m) = o(\log\log m)$ as $m \to \infty$.

A more general result was obtained in 1973 by Nakata [54]: (5.35) remains valid for $\omega(m) = (L_1(m))^{1/2} L_2(m) \ldots L_s(m)$ where s is a natural number, and $L_1(m) = \log m$, $L_s(m) = L_1(L_{s-1}(m))$.

As regards the Walsh system $\{\psi_m(t)\}$ here we proved the first result [36] as early as 1960, namely that if $\omega(m) \equiv 1$, then (see also [38], 64–65)

$$\{\psi_m(t)\} \notin W(uc,\ [0,\ 1],\ \omega(m)). \tag{5.36}$$

The next step in this direction was taken in 1966 by Tandori [52]: (5.36) holds for any $\omega(m) = o(\log\log m)$ as $m \to \infty$. Then in 1978 Nakata [55] obtained a more general result: (5.36) remains valid for all $\omega(m)$ that do not satisfy (5.34). In other words, (5.34) is necessary, just as for the Haar system, for $\omega(m)$ to be a Weyl multiplier for unconditional convergence a.e. of series in the Walsh system. This result was proved somewhat later in 1979 by Bochkarev [56], who used a different method. Next, my student Poleshchuk [51] has shown that these last results also follow from [40].

We draw attention to the fact that the only known sufficient conditions for Weyl multipliers for unconditional convergence of series in the systems $\{\cos mt,\ \sin mt\}$ and $\{\psi_m(t)\}$ are those given by a theorem of Orlicz for general orthogonal series (see [9], 170–172).

Poleshchuk has established a complete analogue of Theorem 5.9 for the Franklin system (see [9], 122–125). He has shown [50] that if $\omega(m) \uparrow \infty$ and $\omega(m) = O(\log^2 m)$ as $m \to \infty$, then there exists an ONS $\{\varphi_m\}$ for which $\omega(m)$ is an exact Weyl multiplier for unconditional convergence a.e.

Remark 5.4. a) The behaviour of the series

$$\sum_{m=2}^{\infty} \frac{\cos mt}{\sqrt{m}\,(\log m)^{\alpha}} \quad \text{for } \frac{1}{2}<\alpha \leqslant \frac{3}{2}$$

as regards unconditional convergence a.e. on $[0, 2\pi]$ is still an open question (see [66], 37-38).

b) It is easy to see that if a function $f \in L(0, 1)$ vanishes on the interval $[a, b] \subset [0, 1]$ then its Fourier-Haar series converges unconditionally a.e. on $[a, b]$, that is unconditional convergence a.e. is a local property of Fourier-Haar series of class L. We do not know for what fairly general non-trivial classes of functions unconditional convergence a.e. of trigonometric Fourier series is also a local property. In particular, there is still no answer for $C(0, 2\pi)$.

Of course, this problem can be considered for other specific systems of functions.

c) It is almost obvious that if an orthogonal series

$$\sum_{m=1}^{\infty} a_m \varphi_m(t) \text{ with } t \in [0, 1] \text{ and } \{a_m\} \in l_2 \tag{5.37}$$

converges a.e. on $[0, 1]$ to some function $F(t)$ after some rearrangement of the terms, then F does not depend on the rearrangement.

On the other hand, the series

$$\sum^{\infty} \frac{\cos 2^m t}{\sqrt{m}} \tag{5.38}$$

does not converge for any order of the terms nor on any set E with $|E| > 0$. However, at almost every point $t_0 \in [0, 2\pi]$ the sum of the positive (negative) terms of the series is $+\infty$ $(-\infty)$.

Thus, the set of sums of all convergent rearrangements of series (5.37) always consists of a single function, but for series (5.38) it is empty (see [62] and [7]).

Another question arises: suppose that we are given a series of functions (orthogonal, trigonometric, Fourier series, and so on)

$$\sum_{m=1}^{\infty} f_m(t),$$

and that it converges almost everywhere (in measure, in L^p, in norm, and so on) under two rearrangements to two different functions F and Φ. What can we say about the collection of all sums? We do not even know whether it must be infinite. Clearly, this question is directly related to Riemann's theorem in function spaces (for this, see Kadets [63], Nikishin [64], and Kornilov [65]).

§6. The Kolmogorov-Men'shov theorem on divergent Fourier series in an orthonormal system of sign functions

The main result, proved in [6], is as follows.

***Theorem* 6.1.** *There exists a system of functions* $\{\varphi_n(t)\}$ *that is orthonormal on* $(0, 1)$ *and takes only the values* ± 1, *and a sequence of numbers* $\{a_n\}$ *for which*

$$\sum_{n=1}^{\infty} a_n^2 < \infty, \tag{6.1}$$

so that the series

$$\sum_{n=1}^{\infty} a_n \varphi_n(t) \tag{6.2}$$

diverges everywhere on $(0, 1)$.

This theorem was proved in 1926. It gave the first example of a uniformly bounded ONS that is not a system of convergence almost everywhere. Moreover, it was established in [6] that Theorem 6.1 remains valid when (6.1) is replaced by

$$\sum_{n=1}^{\infty} a_n^2 \omega(n) < \infty, \tag{6.3}$$

where $\omega(n) = o(\log n)$ as $n \to \infty$.

Subsequent generalizations went in various directions. In 1938 Men'shov [69] proved the following result.

***Theorem* 6.2.** *For any number* $B > 1$ *there is an ON system* $\{\varphi_n(t)\}$ *on* $[0, 1]$ *for which the sequence* $\{\log^2 n\}$ *is an exact Weyl multiplier and* $|\varphi_n(t)| \leqslant B$ *for all* $t \in [0, 1]$ *and* $n = 1, 2, \ldots$

It follows from Theorem 6.2 that the whole class of uniformly bounded ON systems has the same exact Weyl multiplier for convergence a.e. as the class of ONS.

Next, in 1960 we proved the following result (see [36] and [38]):

***Theorem* 6.3.** *There exists a function* $F \in L^p(0, 1)$ *for all* $p > 0$ *such that under any Toeplitz summation method* T *the Fourier-Walsh series for* F *can be rearranged so that the resulting series is not* T*-summable at almost every point* $t \in [0, 1]$.

Clearly, this is a consequence of Theorem 5.6 in §5.

Since the Walsh functions $\psi_n(t)$ only take the values ± 1, Theorem 6.3 generalizes Theorem 6.1 in four directions.

a) In Theorem 6.1 we can take for $\{\varphi_n\}$ a certain rearrangement of the Walsh system $\{\psi_n\}$.

b) In Theorem 6.3 the system $\{\psi_n\}$ is *complete* in $L^2(0, 1)$ whereas it is not known whether in Theorem 6.1 the system $\{\varphi_n\}$ can be made complete by adjoining additional functions that take the values ± 1 only.

c) Theorem 6.3 guarantees the existence of a divergent Fourier series of *any class* L^p, whereas Theorem 6.1 is concerned only with Fourier series of class L^2.

d) In theorem 6.1 a divergent series is constructed, whereas in Theorem 6.3 series are constructed that are *not summable* by any method T.

On the other hand, in 1975 Kashin [67] proved the following result.

***Theorem* 6.4.** *There exists an ON system of functions* $\{\varphi_n(t)\}_{n=1}^{\infty}$ *on* $[0, 1]$ *such that* $|\varphi_n(t)| = 1$ *for* $t \in [0, 1]$, $n = 1, 2, \ldots$ *and*

$$\{\varphi_n\} \notin W(c, [0, 1], \omega(n)) \tag{6.4}$$

for all $\omega(n) = o(\log^2 n)$ *as* $n \to \infty$.

In other words, the sequence $\{\log^2 n\}$ is an *exact* Weyl multiplier for convergence a.e. of the system $\{\varphi_n\}$.

To prove this we need some preliminary lemmas.

***Lemma* 6.1** (see [69] and [6]). *Let* $\{g_n(t)\}_{n=1}^{N}$ *be a sequence of functions in* $L^2(0, 1)$ *such that there exist numbers* γ_i $(1 \leqslant i \leqslant N-1)$ *with*

$$\left|\int_0^1 g_m(t)\, g_l(t)\, dt\right| \leqslant \gamma_i \quad (1 \leqslant m,\ l \leqslant N,\ |m-l| = i) \ \text{ and } \ \sum_{i=1}^{N-1} \gamma_i < M.$$

Then the functions $g_n(t)$ *can be extended to the interval* $(1, 2M+1]$ *as step functions with* $|g_n(t)| = 1$, *and on the whole interval* $[0, 2M+1]$ *they form an orthogonal system.*

***Lemma* 6.2** (see [67]). *Let* n *and* n_1 *be natural numbers with* $n < n_1$, *and* [illegible] *functions such that* $|\varphi_k(t)| = 1$ *for* $t \in [0, 1]$ *and* $k = 1, 2, \ldots, n_1$. *Then there exists a measurable function* $u(t)$ *such that*

$$|u(t)| = 1 \quad \text{for} \quad t \in [0, 1] \tag{6.5}$$

and

$$\int_0^1 \varphi_i(t)\,(u(t)\,\varphi_i(t))\, dt = 0 \quad \text{for} \quad 1 \leqslant i \leqslant n < j \leqslant n_1.$$

Proof. By $\bar{\varepsilon} = \{\varepsilon_m\}_{m=1}^{n_1}$ we denote an arbitrary vector with the coordinates $\varepsilon_k = \pm 1$. We define the sets

$$E(\bar{\varepsilon}) = \{t\colon t \in [0, 1],\ \varphi_m(t) = \varepsilon_m \ \text{ for } \ 1 \leqslant m \leqslant n_1\}.$$

For distinct $\bar{\varepsilon}$ these sets are disjoint and their union is the whole interval $[0, 1]$. It is clear from the definition of $E(\bar{\varepsilon})$ that for a fixed $\bar{\varepsilon}$ each of the functions $\varphi_m(t)$ $(1 \leqslant m \leqslant n_1)$ is constant on $E(\bar{\varepsilon})$. Therefore, if we take $u(t)$ such that

$$|u(t)| \equiv 1 \ \text{ and } \int_{E(\bar{\varepsilon})} u(t)\, dt = 0 \ \text{ for all } \bar{\varepsilon},$$

then (6.5) is satisfied. This proves Lemma 6.2.

Lemma 6.3. *If $p > p_0$, where p_0 is a sufficiently large natural number, then there exists an ONS of step functions $\varphi_1(t), \ldots, \varphi_{2p^2}(t)$ on $[0, 1]$ with the properties*

$$|\varphi_n(t)| = 1 \text{ for } t \in [0, 1] \text{ and } n = 1, 2, \ldots, 2p^2, \tag{6.6}$$

$$m\{t\colon t \in [0, 1], \max_{1 \leqslant m \leqslant 2p^2} \left|\sum_{n=1}^{m} \varphi_n(t)\right| \geqslant C_1 p \log p\} \geqslant C_2, \tag{6.7}$$

where C_1 and C_2 are absolute positive constants.[1]

This lemma is contained in Kashin's paper [67]. A simpler proof which we present here was given by Tandori [68].

Proof of Lemma 6.3. Suppose that (see [9], 166)

$$f_n(t) = \frac{1}{2\left(q-p-n-\frac{1}{2}\right)} \quad \text{for} \quad t \in \left(\frac{q-1}{p}, \frac{q}{p}\right], \quad 1 \leqslant q \leqslant 4p, \quad 1 \leqslant n \leqslant 2p$$

and

$$\alpha_{i,j} = \int_0^4 f_i(t) f_j(t)\, dt.$$

Clearly,

$$\alpha_{i,i} = \frac{1}{4p} \sum_{q=1}^{4p} \frac{1}{\left(q-p-i-\frac{1}{2}\right)^2} \leqslant \frac{2}{p} \sum_{k=1}^{\infty} \frac{1}{(2k-1)^2} = \frac{C_3}{p}, \tag{6.8}$$

and for $i > j$

$$\alpha_{i,j} = \frac{1}{4p} \sum_{q=1}^{4p} \frac{1}{\left(q-p-i-\frac{1}{2}\right)\left(q-p-j-\frac{1}{2}\right)} =$$

$$= \frac{1}{4p(i-j)} \sum_{q=1}^{4p} \left\{\frac{1}{q-p-i-\frac{1}{2}} - \frac{1}{q-p-j-\frac{1}{2}}\right\} =$$

$$= \frac{1}{4p(i-j)} \left\{\sum_{n=1-p-i}^{3p-i} \frac{1}{n-\frac{1}{2}} - \sum_{n=1-p-j}^{3p-j} \frac{1}{n-\frac{1}{2}}\right\} =$$

$$= \frac{1}{4p(i-j)} \left\{\sum_{n=1-p-i}^{-p-j} \frac{1}{n-\frac{1}{2}} - \sum_{n=3p-i+1}^{3p-j} \frac{1}{n-\frac{1}{2}}\right\}.$$

Therefore,

$$|\alpha_{i,j}| \leqslant \frac{1}{4p(i-j)} \left\{\frac{i-j}{p+j+\frac{1}{2}} + \frac{i-j}{3p-i+\frac{1}{2}}\right\} \leqslant \frac{C_4}{p^2} \tag{6.9}$$

$$\text{for} \quad 1 \leqslant i, j \leqslant 2p, \ i \neq j.$$

Suppose that

$$t \in (2, 3], \text{ that is, } t \in \left(\frac{2p+\nu-1}{p}, \frac{2p+\nu}{p}\right] \text{ with } 1 \leqslant \nu \leqslant p.$$

[1] By C_i we denote absolute positive constants.

Then at this point

$$\sum_{n=1}^{p+\nu-1} f_n(t)=\frac{1}{2}\sum_{n=1}^{p+\nu-1}\frac{1}{2p+\nu-p-n-\frac{1}{2}}=$$

$$=\frac{1}{2}\sum_{n=1}^{p+\nu-1}\frac{1}{p+\nu-n-\frac{1}{2}}=\frac{1}{2}\sum_{m=1}^{p+\nu-1}\frac{1}{m-\frac{1}{2}}\geqslant C_5\log p.$$

Therefore,

$$\max_{1\leqslant m\leqslant 2p}\sum_{n=1}^{m} f_n(t)\geqslant C_5\log p \quad \text{for} \quad t\in(2,\,3]. \tag{6.10}$$

Now we set

$$g_{r+(s-1)p}(t)=f_s(t) \quad \text{for} \quad t\in(0,\,4],\ 1\leqslant r\leqslant p \quad \text{and} \quad 1\leqslant s\leqslant 2p$$

and

$$\gamma_i=C_3/p \quad \text{for} \quad 1\leqslant i\leqslant p-1 \quad \text{and} \quad \gamma_i=C_4/p^2 \quad \text{for} \quad p\leqslant i\leqslant 2p^2-1.$$

Obviously,

$$\sum_{i=1}^{2p^2-1}\gamma_i=C_3\frac{p-1}{p}+C_4\frac{2p^2-p}{p^2}\leqslant C_6.$$

Since the $f_n(t)$ are step functions and $|f_n(t)|\leqslant 1$ on $(0,\,4]$, by Lemma 6.1 (see also (6.8) and (6.9)) the functions $g_n(t)$ can be extended to the interval $(4,\,2C_6+4)$ so that the $g_n(t)$ are orthogonal step functions on the whole interval $(0,\,2C_6+4)$ and $|g_n(t)|\leqslant 1$.

Let

$$h_n(t)=g_n((2C_6+4)t) \quad \text{for} \quad t\in(0,\,1] \quad \text{and} \quad 1\leqslant n\leqslant 2p^2.$$

Clearly, these are step functions and orthogonal on $(0,\,1]$. Also, (see (6.10)),

$$|h_n(t)|\leqslant 1 \quad \text{for} \quad t\in(0,\,1] \tag{6.11}$$

$$\text{and} \quad m\{t:\ t\in(0,\,1],\ \max_{1\leqslant m\leqslant 2p^2}\Big|\sum_{n=1}^{m} h_n(t)\Big|\geqslant C_5 p\log p\}\geqslant C_7,$$

because among the g_n each function f_s is repeated p times.

Let $I_1, I_2, \ldots, I_R$ be disjoint intervals in $(0,\,1]$ such that on each I_r the function $h_n(t)$ is constant. We fix an r $(1\leqslant r\leqslant R)$ and put $I_r=(a_r,\,b_r]$, $h_n(t)=\rho_n^{(r)}$, $(t\in I_r,\ 1\leqslant n\leqslant 2p^2)$. In $(0,\,1]$ we define independent step functions (in the sense of probability) $\chi_n^{(r)}$ $(1\leqslant n\leqslant 2p^2)$ that are orthogonal and take only the two values[1] $1-\rho_n^{(r)}$ and $-1-\rho_n^{(r)}$ and are such that

$$\int_0^1 \chi_n^{(r)}(t)\,dt=0 \quad \text{for all} \quad 1\leqslant n\leqslant 2p^2.$$

We put

$$\chi_n^{(r)}(I_r;\ t)=\begin{cases}\chi_n^{(r)}\left(\dfrac{t-a_r}{b_r-a_r}\right) & \text{for} \quad t\in I_r,\\ 0 & \text{for} \quad t\notin I_r,\end{cases}$$

[1] If $|\rho_n^{(r)}|=1$ we must set $\chi_n^{(r)}(t)\equiv 0$.

and finally,

$$(6.12)\qquad \varphi_n(t)=h_n(t)+\sum_{r=1}^{R}\chi_n^{(r)}(I_r;\ t)\quad \text{for}\quad t\in[0,\ 1]\ \text{and}\ 1\leqslant n\leqslant 2p^2.$$

From the definition we see that $|\varphi_n(t)|\equiv 1$ for $t\in(0,1)$ and that for $i\neq j$

$$\int_0^1 \varphi_i(t)\,\varphi_j(t)\,dt=\int_0^1 h_i(t)\,h_j(t)\,dt+\sum_{r=1}^{R}\int_{I_r} h_i(t)\,\chi_j^{(r)}(I_r;\ t)\,dt+$$

$$+\sum_{r=1}^{R}\int_{I_r} h_j(t)\,\chi_i^{(r)}(I_r;\ t)\,dt+\sum_{r=1}^{R}\int_{I_r}\chi_i^{(r)}(I_r;\ t)\,\chi_j^{(r)}(I_r;t)\,dt=$$

$$=\sum_{r=1}^{R}\rho_i^{(r)}mI_r\int_0^1\chi_j^{(r)}(t)\,dt+\sum_{r=1}^{R}\rho_j^{(r)}mI_r\int_0^1\chi_i^{(r)}(t)\,dt+\sum_{r=1}^{R}mI_r\int_0^1\chi_i^{(r)}(t)\,\chi_j^{(r)}(t)\,dt=0.$$

Thus, the functions $\varphi_n(t)$ $(1\leqslant n\leqslant 2p^2)$ form an ONS on $(0,1)$.[1] Hence, (6.6) holds.

We now choose an interval I_r for which

$$(6.13)\qquad \max_{1\leqslant m\leqslant 2p^2}\Big|\sum_{n=1}^{m}h_n(t)\Big|\geqslant C_5 p\log p\quad \text{for}\quad t\in I_r.$$

Since the $\chi_n^{(r)}$ are independent and $|\chi_n^{(r)}(t)|\leqslant 2$ (see (6.11)), by Kolmogorov's inequality ([72], vol. 1, 247-248 or [73], 371-372)

$$(6.14)\qquad m\Big\{t:\ t\in I_r,\ \max_{1\leqslant m\leqslant 2p^2}\Big|\sum_{n=1}^{m}\chi_n^{(r)}(I_r;\ t)\Big|\geqslant C_5\frac{p\log p}{2}\Big\}\leqslant$$

$$\leqslant\sum_{n=1}^{2p^2}\Big(\frac{2}{C_5 p\log p}\Big)^2\int_{I_r}[\chi_n^{(r)}(I_r;\ t)]^2\,dt\leqslant C_8\frac{mI_p}{\log^2 p}.$$

We take $p>p_0$, where $\log^2 p_0>2C_8$. Then by (6.12)-(6.14)

$$m\Big\{t:\ t\in I_r,\ \max_{1\leqslant m\leqslant 2p^2}\Big|\sum_{n=1}^{m}\varphi_n(t)\Big|\geqslant C_5\frac{p\log p}{2}\Big\}\geqslant\frac{mI_r}{2}\quad \text{for}\quad p>p_0.$$

Therefore (see (6.11)),

$$m\Big\{t:\ t\in(0,\ 1),\ \max_{1\leqslant m\leqslant 2p^2}\Big|\sum_{n=1}^{m}\varphi_n(t)\Big|\geqslant C_5\frac{p\log p}{2}\Big\}\geqslant\frac{C_7}{2},$$

that is, (6.7) holds, and the lemma is proved.

Proof of Theorem 6.4 (see [67], 263-264). We take all odd integers $q>p_0$ and in accordance with Lemma 6.3 with $p=2^{(q-1)/2}$ we construct groups of functions

$$\varphi_n^q(t)\quad \text{for}\quad 1\leqslant n\leqslant 2p^2=2^q\quad \text{and}\quad q>p_0.$$

For the remaining q, that is, $0\leqslant q\leqslant p_0$, or for even q we construct groups of functions $\varphi_n^q(t)$ with $1\leqslant n\leqslant 2^q$ from any pairwise distinct Rademacher

[1] That they are normalized is obvious, since $|\varphi_n(t)|\equiv 1$.

functions (see [9], 125). By Lemma 6.2, we can find by induction functions $u_q(t)$ with $u_0(t) \equiv 1$ and $|u_q(t)| \equiv 1$ for $q \geqslant 1$, such that the system

$$\{u_q(t)\,\varphi_n^q(t)\} \quad \text{with} \quad 1 \leqslant n \leqslant 2^q \quad \text{and} \quad 0 \leqslant q < \infty$$

is ON on the interval $[0, 1]$.

We put

$$(6.15) \qquad \varphi_m(t) = u_q(t)\,\varphi_n^q(t) \text{ for } m = 2^q + n \quad \text{with } 1 \leqslant n \leqslant 2^q \text{ and } q = 0, 1, \ldots$$

The system $\{\varphi_m\}_{m=1}^{\infty}$ is what we are looking for. For suppose that we are given a positive sequence $\{\omega(m)\}$ with $\omega(m)\uparrow$ and $\omega(m) = o(\log^2 m)$ as $m \to \infty$. Then we can find odd numbers q_j such that

$$(6.16) \qquad 0 < \omega(2^{q_j+1}) \leqslant \frac{1}{j^4}\left(\frac{q_j-1}{2}\right)^2, \quad p_0 < q_1 < q_2 < \ldots \quad (j = 1, 2, \ldots),$$

and we put

$$(6.17) \qquad a_m = \begin{cases} \dfrac{1}{j}\,(2^{q_j-1}\omega(2^{q_j+1}))^{-1/2} & \text{for} \quad 2^{q_j} < m \leqslant 2^{q_j+1} \quad \text{and} \quad j \geqslant 1, \\ 0 & \text{for the remaining } m. \end{cases}$$

Firstly, it follows from (6.17) that

$$(6.18) \qquad \sum_{m=1}^{\infty} a_m^2\omega(m) = \sum_{j=1}^{\infty} \sum_{m=2^{q_j}+1}^{2^{q_j+1}} a_m^2\omega(m) \leqslant$$

$$\leqslant \sum^{\infty} \frac{1}{j^2}\,\frac{\omega(2^{q_j+1})\,2^{q_j}}{\omega(2^{q_j-1})\,\omega(2^{q_j+1})} = \sum^{\infty} \frac{2}{j^2} < \infty,$$

that is, a condition of the form (6.3) is satisfied. Secondly, at t_0 we have

$$\max_{1 \leqslant m \leqslant 2^{q_j}} \Big|\sum_{n=1}^{m} \varphi_n^{q_j}(t_0)\Big| \geqslant C_1 p \log p \quad \text{with} \quad p = 2^{\frac{q_j-1}{2}}.$$

By (6.15)-(6.17) this means that a fortiori at t_0

$$\max_{1 \leqslant \nu \leqslant 2^{q_j+1}} \Big|\sum_{m=2^{q_j}+1}^{\nu} a_m\varphi_m(t_0)\Big| \geqslant C_1 j,$$

therefore (see (6.7))

$$m\,\{t\colon t \in (0, 1), \quad \max_{1 \leqslant \nu \leqslant 2^{q_j+1}} \Big|\sum_{m=2^{q_j}+1}^{\nu} a_m\varphi_m(t)\Big| \geqslant C_1 j\} \geqslant C_2 \quad (j = 1, 2, \ldots).$$

From this estimate it follows that (6.2) diverges to infinity on some set of positive measure. If we now take account of (6.18), we find that (6.4) is satisfied. This completes the proof of Theorem 6.4.

Remark 6.1. It is not known whether in Theorem 6.4 we can construct a series (6.2) that diverges almost everywhere and make the system $\{\varphi_m\}$ complete.

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